# Strategic Under-Specification: A Stackelberg Game Between User and Assistant

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**Abstract**

Users of coding assistants routinely write terse prompts, relying on the assistant to resolve missing details through guessing or clarifying questions. We develop a contract-theoretic model of endogenous clarification in human–AI interaction. Prompt shortening reflects rational effort substitution ($m^*(a) \leq m^*(0)$) rather than inefficient moral hazard: holding the clarification policy fixed, the envelope theorem implies that better autonomous guessing weakly improves user welfare even as prompts shorten. When clarification policies are opaque, naive users who overestimate the assistant's willingness to ask under-specify relative to what the true policy requires, incurring welfare losses that are quadratic in the belief error. Conversely, under transparent Stackelberg commitment, an assistant with asking friction bias ($\lambda_A > c_Q$) suppresses clarification, inducing sophisticated users to defensively over-specify upstream ($m^* > m^{FB}$) to insure against unaddressed defects.

Closing the loop under uniform *pooling* and effort-conditioned *screening* governance regimes, we characterize equilibrium clarification and welfare. Under pooling, provider asking friction suppresses clarification, generating an equilibrium capability reversal (the "guessing trap"): when the assistant's asking rate declines sufficiently steeply with autonomous accuracy, prompt lengths increase and user welfare falls. Under screening menus, provider bias induces adverse-selection downward distortion on high-cost terse users ($a_H^{SB} < a_H^B$), bounded above by the relative bias $\tilde{\rho}/s$, while boundary relaxation moderates this distortion. Because there are no transfers, in the saturation regime the assistant deters mimicry by degrading the terse user's bundle rather than paying the efficient specifier an information rent, which reverses the classical rent pattern. The menu is implementable even when users may type any prompt. Under mild bias, pooling can deliver strictly higher user welfare than screening, because a biased screening assistant exploits contractible specification to dictate inefficiently high prompt effort. We separate genuine adverse-selection distortion from technological clarification shutdown, state when the unclipped closed forms hold and when boundary clipping binds, and propose a pre-registrable experimental design with a power analysis for human-subject testing.

**Keywords:** Human–AI Interaction, Clarification Policies, Stackelberg Games, Contract Theory, Adverse Selection, Delegation.

**Code Availability:** Simulation suite and verification scripts are available at `https://github.com/pranjal-0802/strategic-underspecification`.

**AI-Use Disclosure:** AI assistance tools were used for simulation code development, test suite execution, and copy-editing suggestions. All conceptual formulations, economic models, mathematical proofs, and analytical interpretations were developed, derived, and verified by the author.

## 1 Introduction

A user typing a coding request faces a fundamental economic trade-off: spend cognitive effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the

assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user's specification effort depends on what they expect the assistant to do with silence, and the assistant's ask/guess policy is chosen (explicitly, or as an emergent property of training and product incentives) with some implicit model of how users will respond.

This paper provides a mechanism-design theory of endogenous clarification in AI assistance. Its central premise is that prompt shortening reflects rational effort substitution in response to automated clarification ($m^*(a) \leq m^*(0)$), while welfare-reducing under-specification arises when users naively anticipate more clarification than the assistant provides ($m^*_{\text{naive}} < m^*_{\text{soph}}$). When the assistant's policy is transparently chosen as a Stackelberg leader, provider asking friction ($\lambda_A > c_Q$) suppresses clarification, forcing users to over-specify relative to first-best ($m^* > m^{FB}$) and generating an equilibrium capability trap where better guessing can perversely lengthen prompts and lower welfare. Section 2 begins with a decision-theoretic benchmark where the assistant's policy is exogenous, proving that improved guessing accuracy alone weakly improves user welfare despite shortening prompts. Section 3 formalizes effort substitution and belief-driven under-specification under opaque policies. Sections 4–7 close the loop by modeling the assistant as a Stackelberg leader anticipating user best responses under pooling and screening instruments. Section 8 translates the equilibrium characterizations into falsifiable behavioral predictions, and Section 9 proposes an experimental design to test them. Section 10 evaluates exact nonlinear robustness and discusses institutional extensions.

### 1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

**Contract theory and screening without transfers.** Our screening framework builds on classical adverse-selection and mechanism design models (Mirrlees, 1971; Myerson, 1979, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate $a$ offered in response to specification effort $m$). This connects to non-monetary screening and costly verification (Townsend, 1979; Ben-Porath et al., 2014), where information revelation is governed by operational friction rather than cash transfers.

**Delegation, authority, and non-monetary mechanism design.** Our analysis relates closely to delegation and non-monetary mechanism design (Holmström, 1984; Dessein, 2002; Alonso and Matouschek, 2008; Armstrong and Vickers, 2010; Ambrus and Egorov, 2017), where a principal designs an optimal decision menu without monetary transfers, and to models of formal versus real authority (Aghion and Tirole, 1997), where an agent's upstream effort is shaped by who retains control over downstream execution. In our setting, leaving attributes unspecified implicitly delegates their resolution to the assistant. Furthermore, because the assistant's asking policy determines which state dimensions are actively resolved, our framework shares conceptual foundations with Bayesian persuasion and strategic information design (Kamenica and Gentzkow, 2011), while focusing on the effort-substitution trade-off in prompt articulation.

**Costly communication and strategic disclosure.** The user's specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user's rational economizing on specification costs when anticipating automated assistance.

**Assistance games and human–AI interaction.** Our bi-level formulation models the interaction as a Stackelberg game (von Stackelberg, 1934), connecting conceptually to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016), as well as incomplete contracting in AI alignment (Hadfield-Menell and Hadfield, 2019), where an autonomous assistant acts to maximize human utility under incomplete specification. However, while assistance games formalize the assistant's uncertainty over the human's underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value $V$ and error loss $L$). The user privately knows the attribute values $\theta$, but these are revealed truthfully when specified or asked about; the asymmetry that the assistant's policy must screen is the user's private *specification cost* $\kappa$. The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. Our model provides a contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

**Clarification, ambiguity, and RLHF incentives.** A parallel literature in natural language processing investigates algorithms for selective clarification under ambiguity (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020; Min et al., 2020; Kuhn et al., 2023). Recent evaluations show that state-of-the-art language models often guess prematurely under ambiguity rather than asking clarifying questions (Laban et al., 2025). In conversational AI, this premature guessing is reinforced by reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO), which systematically favor immediate answers and introduce conversational length and sycophancy biases (Sharma et al., 2023; Singhal et al., 2023). Existing engineering studies treat prompts as exogenous static benchmarks. Our framework endogenizes prompt specification effort as an equilibrium best response to provider clarification policies and asking frictions.

# 2 Environment and the Decision-Theoretic Benchmark

## 2.1 Setup

A task has $k$ binary, independent, intent-relevant attributes $\theta = (\theta_1, \ldots, \theta_k) \in \{0,1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in \{0, \ldots, k\}$ and truthfully reveals $m$ attributes.[1] Specifying $m$ attributes costs $c(m)$, with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$.

For each of the $k-m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0,1)$. An uninformed guess on a uniform binary attribute succeeds with probability $1/2$; $g > 1/2$ reflects context the assistant can exploit (repository conventions, priors over common choices), and $g < 1/2$ arises

[1] We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

for non-binary attributes. We model the assistant's policy at this stage as a uniform ask rate $a \in [0,1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a,g) = a + (1-a)g$. Task success requires *all* $k$ attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user's payoff is

$$U(m; a, g) \;=\; V\, q(a,g)^{\,k-m} \;-\; c(m) \;-\; a\, c_Q\,(k-m),$$

with $V > 0$ the value of success and $c_Q > 0$ the user's cognitive cost of answering one clarifying question. We treat $m$ as continuous on $[0,k]$ for tractability.

### 2.2 Proposition 1: capability improvements and effort substitution

**Proposition 1.** *Fix $a$. Let $U^*(g) = \max_{m\in[0,k]} U(m;a,g)$. Then $U^*$ is weakly increasing in $g$.*

*Proof.* For any fixed $m \in [0,k]$, the user's objective $U(m;a,g) = Vq(a,g)^{k-m} - c(m) - ac_Q(k-m)$ is pointwise non-decreasing in $g$ because $\partial U/\partial g = V(k-m)q(a,g)^{k-m-1}(1-a) \geq 0$. As the upper envelope of a family of weakly increasing functions, the maximized utility $U^*(g) = \max_{m\in[0,k]} U(m;a,g)$ is weakly increasing in $g$ directly, without requiring differentiability. Under differentiability, the envelope theorem confirms that at the interior optimum $m^*(g)$,

$$\frac{dU^*}{dg} \;=\; \left.\frac{\partial U}{\partial g}\right|_{m=m^*(g)} \;=\; V(k-m^*)q^{k-m^*-1}(1-a) \;\geq\; 0.$$

The envelope theorem lets us ignore $dm^*/dg$: because $m^*$ is chosen optimally, its first-order response to $g$ has no first-order effect on $U^*$. □

**Assumption 1** (Regularity)**.** *In the relevant range of $m$, $(k-m)\ln q(a,g) \geq -1$.*

**Remark 1** (Interpreting Assumption 1 and Model Concavity)**.** *The regularity condition $(k-m)\ln q(a,g) \geq -1$ provides a sufficient condition for decreasing differences in $(m,g)$, which guarantees monotone comparative statics via Topkis's theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994). The direction is pinned down by the cross-partial of utility: for fixed $a$,*

$$\frac{\partial^2 U}{\partial m\,\partial g} \;=\; -V(1-a)\,q^{\,k-m-1}\Big[(k-m)\ln q + 1\Big].$$

*Thus $U$ exhibits decreasing differences in $(m,g)$ whenever $(k-m)\ln q + 1 \geq 0$, which is precisely Assumption 1. Because this condition is most demanding at $m=0$, decreasing differences on the entire domain $[0,k]$ (the hypothesis of Topkis's theorem) requires $k\ln q \geq -1$. When $k$ is large or baseline accuracy is modest (e.g., $k=10, g=0.5$), this global condition is violated. A weaker local statement is available: at an interior optimum $m^*$ satisfying the second-order condition $\partial^2 U/\partial m^2 < 0$, the implicit function theorem gives $dm^*/dg = -U_{mg}/U_{mm}$, whose sign equals that of $U_{mg}$ evaluated at $m^*$. The local condition $(k-m^*)\ln q \geq -1$ therefore suffices for $dm^*/dg \leq 0$ along a smooth branch of optima. Users who articulate most attributes upstream keep $k-m^*$ small, and the local condition holds in 89.9% of our grid configurations (Section 10). Where it fails, $U$ exhibits increasing differences at $m^*$, and higher guessing accuracy induces* higher *specification effort: monotonicity breaks in 96.7% of the grid intervals where the local condition fails, so the condition is close to necessary as well as sufficient.*

*We also note that in the exact conjunctive model, the gross benefit $Vq^{k-m}$ is strictly convex in $m$ (since $\partial^2[Vq^{k-m}]/\partial m^2 = Vq^{k-m}(\ln q)^2 > 0$). Under quadratic cost $c(m) = \kappa m^2/2$, $U$ is strictly concave in $m$ on $[0,k]$ if and only if $\kappa \geq V(\ln q)^2$ (the supremum of $Vq^{k-m}(\ln q)^2$, attained*

*at $m = k$); otherwise the optimum may jump between local maxima, and the local argument above applies only along each smooth branch. Under the linear-risk specification introduced in Section 4, by contrast, gross benefit is linear in $m$, guaranteeing that $U$ is globally strictly concave in $m$ for all $\kappa > 0$.*

**Corollary 1.** *If Assumption 1 holds for all $m \in [0, k]$ and all $g$ in the range considered, the set of maximizers $m^*(g)$ is weakly decreasing in $g$ in the strong set order (Topkis's theorem). If it holds only at an interior optimum satisfying the second-order condition, $m^*(g)$ is locally weakly decreasing along that branch.*

*Proof.* Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition $\partial^2 U / \partial m \, \partial g \leq 0$. On all of $[0, k]$ this is decreasing differences in $(m, g)$, and Topkis's theorem applies; at a single interior optimum, the implicit function theorem gives $dm^*/dg = -U_{mg}/U_{mm} \leq 0$. □

Thus, holding the assistant's policy fixed, users specify less as the assistant becomes more adept at guessing, yet they are never worse off for doing so. Proposition 1 serves primarily as a baseline: it applies the envelope theorem to a single agent optimizing against a technology parameter that enters the payoff with a non-negative derivative, a property that holds generally in models of this class. Its purpose is not to present a surprising result, but to establish the rigorous benchmark against which claimed welfare loss from prompt shortening must be evaluated, ensuring that Proposition 2 isolates the true source of strategic inefficiency. This frames the central question of this paper: what happens once the policy $a$ is no longer exogenous, but chosen by a strategic assistant that anticipates the user's response?

# 3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant's ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

## 3.1 Cost externality

If a third party (such as a code reviewer or downstream production system) bears part of the cost of an erroneous guess, the user's private defect loss $L$ understates the social loss, causing the user to under-specify relative to the social optimum at any fixed $(a, g)$. Under the linear-risk specification of Section 4 with social defect loss $L_{\text{soc}} > L$, the gap between social and private specification effort is $(L_{\text{soc}} - L)(1 - g)(1 - a)/\kappa$ for interior solutions. The gap contracts as guessing accuracy $g$ improves, because autonomous errors become less frequent, and vanishes under full clarification ($a = 1$). While this constitutes a standard Pigouvian externality, our primary focus lies in strategic and informational friction.

## 3.2 Miscalibrated belief updating under silent failure

If incorrect guesses are not always detected, and the detection probability $d(m)$ is increasing in $m$, users infer $g$ from experience rather than observing it directly. Under-specifying users are consequently less likely to catch their own errors, generating an upward-biased estimate $\hat{g}$ that reinforces under-specification. This mechanism represents a dynamic learning process rather than a static equilibrium outcome; we formalize its testable implications in Section 8 (prediction P5).

### 3.3 Belief about the assistant's policy and equilibrium under-specification

When the clarification policy $a$ is endogenous, the user's specification decision depends on their belief regarding the assistant's asking rate. Let users be *sophisticated* (correctly inferring the assistant's true operational ask rate $a_A$) or *naive* (believing the assistant follows a higher clarification rate $a^\dagger > a_A$, such as the unconstrained welfare-optimal policy).

**Definition 1** (Effort Substitution and Belief-Driven Under-Specification)**.** *A user interacting with an automated assistant modifies upstream specification effort $m$ across two distinct economic margins:*

*(i)* ***Rational effort substitution:*** *$m^*(a) \leq m^*(0)$, where the user economizes on prompt articulation by delegating attribute resolution to automated assistance; and*

*(ii)* ***Belief-driven under-specification:*** *$m^*(\hat{a}) < m^*(a)$, where a user holding optimistic or miscalibrated beliefs $\hat{a} > a$ under-specifies relative to the assistant's true operational ask rate.*

In human–AI interaction, varying the clarification rate $a$ engages two opposing economic channels:

1. *The accuracy channel:* A higher ask rate increases effective resolution accuracy $q(a, g) = a + (1 - a)g$. Under decreasing differences (Assumption 1), greater accuracy reduces the marginal value of upstream prompt effort, pushing specification downward.

2. *The question-cost channel:* Each clarifying question imposes an interactive answering cost $c_Q$ on the user. Because the total answering cost $ac_Q(k - m)$ is decreasing in specification $m$, a higher ask rate increases the marginal penalty of leaving attributes unspecified, pushing specification upward (pre-emption).

Under the linear-risk specification analyzed in Sections 4–6, where the net benefit of clarification $s \equiv \Lambda - c_Q > 0$ is positive, the cross-partial derivative of user utility is:

$$\frac{\partial^2 U}{\partial m \, \partial a} \;=\; -(\Lambda - c_Q) \;<\; 0.$$

The accuracy channel strictly dominates the question-cost channel, ensuring that $U$ exhibits strictly decreasing differences in $(m, a)$.

**Proposition 2** (Prompt Under-Specification under Naive Beliefs)**.** *Under the linear-risk specification with $\Lambda > c_Q$, naive users choose weakly lower specification effort than sophisticated users:*

$$m^*_{naive} \;\leq\; m^*_{soph},$$

*holding strictly ($m^*_{naive} < m^*_{soph}$) for all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$. Consequently, naive users achieve weakly lower true welfare, $U(m^*_{naive}; a_A) \leq U(m^*_{soph}; a_A)$, strictly so on the interior domain. Holding raw guessing accuracy $g$ fixed, naive users under-specify relative to the assistant's true operational policy and suffer an equilibrium welfare loss.*

*Proof.* Under linear risk on the physical domain $m \in [0, k]$, the follower's best response to an anticipated ask rate $a$ is $m^*(\kappa; a) = \min\{k, \beta(\kappa; a)\}$, where $\beta(\kappa; a) \equiv \frac{\Lambda(1-a)+ac_Q}{\kappa}$. Since $\Lambda > c_Q$, the unconstrained response $\beta(\kappa; a)$ is strictly decreasing in $a$: $\partial\beta/\partial a = (c_Q - \Lambda)/\kappa < 0$. A naive user anticipating $a^\dagger > a_A$ therefore satisfies $\beta(\kappa; a^\dagger) < \beta(\kappa; a_A)$. For low-cost types with $\kappa \leq$

$\frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) \geq k$, so $\beta(\kappa; a_A) > k$ as well. Both beliefs therefore yield boundary saturation: $m^*_{\text{naive}} = m^*_{\text{soph}} = k$, and true welfare is identical ($U(m^*_{\text{naive}}; a_A) = U(m^*_{\text{soph}}; a_A)$). For all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) < k$, which guarantees:

$$m^*_{\text{naive}} \;=\; \beta(\kappa; a^\dagger) \;<\; \min\{k, \beta(\kappa; a_A)\} \;=\; m^*_{\text{soph}}.$$

Because true utility $U(m; a_A)$ is strictly concave in $m$ with unique global maximizer on $[0, k]$ at $m^*_{\text{soph}}$, any choice $m \neq m^*_{\text{soph}}$ incurs a strict utility loss. On the fully interior domain where $m^*_{\text{soph}} = \beta(\kappa; a_A) < k$, evaluating the quadratic shortfall directly yields:

$$U(m^*_{\text{soph}}; a_A) - U(m^*_{\text{naive}}; a_A) \;=\; \frac{\kappa}{2}\big(m^*_{\text{soph}} - m^*_{\text{naive}}\big)^2 \;=\; \frac{(\Lambda - c_Q)^2}{2\kappa}\,(a^\dagger - a_A)^2 \;>\; 0,$$

confirming that interior naive users are strictly worse off.

Notice that on the interior domain, the under-specification gap is:

$$m^*_{\text{soph}} - m^*_{\text{naive}} \;=\; \frac{(\Lambda(g) - c_Q)(a^\dagger - a_A)}{\kappa} \;=\; \frac{(L(1-g) - c_Q)(a^\dagger - a_A)}{\kappa}.$$

Differentiating with respect to raw guessing accuracy $g$, holding the policies $(a^\dagger, a_A)$ fixed, yields:

$$\frac{\partial(m^*_{\text{soph}} - m^*_{\text{naive}})}{\partial g} \;=\; -\,\frac{L(a^\dagger - a_A)}{\kappa} \;<\; 0.$$

As the assistant's autonomous guessing capability improves, the distortion gap between naive and sophisticated users strictly contracts, because guessing errors become less frequent and the penalty for misplaced trust declines. (When $a_A$ itself responds to $g$, as in Section 5, the total derivative also includes $-(s/\kappa)\, da_A/dg$, which can have either sign.) □

**Remark 2** (Linear Risk Foundation and Non-Linear Cross-Partial Reversal)**.** *The monotone under-specification result of Proposition 2 relies directly on the linear-risk structure, where the cross-partial derivative is identically $\partial^2 U/\partial m\,\partial a = -(\Lambda - c_Q) < 0$. Under linear risk, whenever clarification is cost-effective ($\Lambda > c_Q$), user utility exhibits decreasing differences globally, ensuring that anticipated asking strictly reduces prompt effort ($m^*(\hat{a}) < m^*(a)$ for $\hat{a} > a$). In contrast, in the exact conjunctive model $U(m, a) = Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the cross-partial derivative is:*

$$\frac{\partial^2 U}{\partial m\,\partial a} \;=\; -V(1-g)\, q^{k-m-1}\big[(k-m)\ln q + 1\big] \;+\; c_Q.$$

*Here, varying the ask rate activates two opposing forces: the accuracy benefit $-V(1-g)q^{k-m-1}[(k-m)\ln q + 1] \leq 0$ and the interactive question-answering friction $+c_Q > 0$. When question costs are large relative to task stakes or when autonomous accuracy is high, the question-cost channel dominates. For example, take $k = 5, V = 10, \kappa = 0.5, g = 0.8, c_Q = 2$, so that the linearized stake $V(1-g) = 2.0 = c_Q$ and the first-order net asking surplus is zero. Asking then has negative net value for the user (optimized utility 4.16 at $a = 0$ versus 4.08 at $a = 0.1$), and a naive user who believes $a^\dagger = 0.6$ while the true policy is $a_A = 0.1$ chooses* higher *upstream effort ($m^*_{naive} = 3.92 > 3.15 = m^*_{soph}$) to pre-empt conversational interruptions. Proposition 2 thus isolates the economic environment where clarification provides positive net value ($\Lambda > c_Q$), which represents the canonical operating regime for helpful automated assistants.*

This identifies the dual nature of specification distortions across informational regimes:

1. *Opaque policies and strategic under-specification:* When users naively trust the assistant to provide first-best clarification ($a^\dagger = a^{FB} = 1$), they write concise prompts calibrated to full assistance ($m^*_{\text{naive}} = m^{FB} = c_Q/\kappa$). If the assistant is a biased Stackelberg leader implementing an asking-suppressed policy $a_A = a^{SE} < 1$, naive users under-specify relative to what the biased environment demands ($m^*_{\text{naive}} < m^*_{\text{soph}}$). Their welfare shortfall from misplaced trust is:

$$\Delta U_{\text{naive}} \;\equiv\; U(m^*_{\text{soph}}; a^{SE}) \;-\; U(m^*_{\text{naive}}; a^{SE}) \;=\; \frac{(\Lambda - c_Q)^2}{2\kappa}\big(1 - a^{SE}\big)^2 \;=\; \frac{s^2(1-a^{SE})^2}{2\kappa},$$

   which is strictly positive whenever the leader strategically withholds clarification ($a^{SE} < 1$).

2. *Transparent policies and defensive over-specification:* Under transparent Stackelberg commitment, sophisticated users know the leader's operational policy $a^{SE}$. Because the biased leader under-asks ($a^{SE} < 1$), sophisticated users cannot rely on clarification downstream. They compensate by defensively lengthening prompts beyond the first-best level ($m^*(a^{SE}) > m^{FB}$), substituting costly private effort for missing automated assistance.

Sections 4–7 close the loop by modeling how the assistant endogenously chooses this clarification policy under pooling and screening governance mechanisms.

# 4 The Assistant's Policy as a Stackelberg Game

## 4.1 General formulation

The assistant (leader) commits ex ante to a policy $\pi$ before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m;\kappa) = \kappa m^2/2$, and best-responds $m^*(\kappa;\pi) = \arg\max_m U(m;\pi,\kappa)$. The leader's own objective $\Pi$ need not equal $U$: we write

$$\Pi_\kappa(m,a) \;=\; \mu_A\, U(m,a;\kappa) \;-\; (\lambda_A - c_Q)\, a\,(k-m),$$

where $\mu_A \in (0,1]$ discounts the leader's weight on true downstream user welfare, $c_Q$ is the user's true cognitive cost of answering one clarifying question, and $\lambda_A$ is the leader's own perceived or institutional cost per clarifying question. In commercial AI systems, misalignment along these margins arises from concrete engineering and economic forces:

1. *Asking friction bias* ($\lambda_A > c_Q$): Reinforcement learning from human feedback (RLHF) and preference optimization algorithms (such as DPO) commonly evaluate conversation turns in isolation. Reward models systematically penalize "evasive" clarifying questions in favor of immediate code generation because rating pools favor direct answers, reinforcing sycophancy and conversational turn penalties (Sharma et al., 2023; Singhal et al., 2023; Laban et al., 2025). Furthermore, multi-turn interactions increase token costs, context-window expansion, and inference latency, creating provider-side economic pressure to suppress clarification turns.

2. *Downstream welfare discounting* ($\mu_A < 1$): Turn-level user satisfaction signals (such as prompt acceptance or immediate positive feedback) capture immediate session fluency. In contrast, the true economic loss $L$ from broken conventions, subtle edge-case errors, or security defects is typically discovered hours or days later in downstream execution, test suites, or code review. Consequently, training objectives tuned to immediate session retention tend to discount downstream correctness.

When asking friction is aligned ($\lambda_A = c_Q$), the leader's objective satisfies $\Pi_\kappa \equiv \mu_A U(\cdot;\kappa)$, so it has the same maximizers as true user welfare for any positive welfare weight $\mu_A > 0$. Because user utility $U$ already incorporates private questioning cost $-ac_Q(k-m)$, expanding the leader's objective yields:

$$\Pi_\kappa(m,a) \;=\; \mu_A\left[V \;-\; \Lambda(1-a)(k-m) \;-\; \frac{\kappa}{2}m^2\right] \;-\; \Big(\lambda_A - (1-\mu_A)c_Q\Big)\,a\,(k-m).$$

The total effective weight multiplying clarification volume $-a(k-m)$ in $\Pi_\kappa$ is therefore $\lambda_A - (1-\mu_A)c_Q$, which simplifies directly to $\lambda_A$ in the benchmark case $\mu_A = 1$. Downstream welfare discounting $\mu_A < 1$ creates policy distortion through interaction with excess asking friction ($\lambda_A \neq c_Q$), inflating the effective friction per unit of altruism $(\lambda_A - c_Q)/\mu_A$. The leader solves $\pi^* = \arg\max_\pi \mathbb{E}_{\kappa\sim F}[\Pi_\kappa(m^*(\kappa;\pi),\pi)]$, a bilevel program.

**Remark 3** (Existence)**.** *For Model I below ($\pi = a \in [0,1]$, a compact scalar policy), $m^*(\kappa;a)$ is continuous in $a$ by Berge's Maximum Theorem (since $U$ is strictly concave in $m$ on the compact domain $[0,k]$), so the expected leader objective $\mathbb{E}_\kappa[\Pi_\kappa(m^*(\kappa;a),a)]$ is continuous on $[0,1]$ and a maximizer exists by Weierstrass's Extreme Value Theorem. For Model II below (a finite menu over two types), the feasible allocation space is a non-empty, compact subset of $[0,k]^2 \times [0,1]^2$ defined by continuous (bilinear in $(m,a)$ and quadratic in $m$) IC and IR constraints. The constraint set is non-empty because the zero-asking menu $\{(m_L^0,0),(m_H^0,0)\}$ with $m_\kappa^0 = \min\{k,\Lambda/\kappa\}$ is feasible: each type receives its own unassisted optimum, so both IC constraints hold, and IR holds for any reservation utility not exceeding the unassisted payoff. By Weierstrass's theorem, the continuous leader objective attains a global maximum on this compact set.*

For tractability in what follows we evaluate the linear-risk objective:

$$U(m,a;\kappa) \;=\; V \;-\; \Lambda(1-a)(k-m) \;-\; \frac{\kappa}{2}m^2 \;-\; a\,c_Q\,(k-m),$$

where $\Lambda \equiv L(1-g)$. This specification admits two complementary economic microfoundations:

1. *Additive independent defects ($kL \le V$):* Each mis-resolved attribute independently produces a localized defect cost $L \le V/k$, so losses add up across attributes and exhaust the task value only if every attribute is defective. Our benchmark calibration sets $k = 10, L = 10, V = 100$, where full failure ($kL = 100$) exactly exhausts task value $V$.

2. *Systemic bottleneck conjunctive model ($L = V$):* The project succeeds only if all attributes are defect-free: $Vq(a,g)^{k-m}$. A first-order Taylor expansion around $q \to 1$ yields $Vq^{k-m} \approx V - V(1-q)(k-m)$, which corresponds to setting per-attribute stake $L = V = 100$.

Throughout Sections 4.2–6, we maintain the physical feasibility constraint $m \in [0,k]$, so that followers cannot specify more attributes than exist in the task. When unconstrained first-order conditions yield $m^*(\kappa;a) = \frac{\Lambda(1-a)+ac_Q}{\kappa} > k$ (which occurs for low-cost types with $\kappa < [\Lambda(1-a) + ac_Q]/k$, a threshold that falls from $\Lambda/k$ at $a = 0$ to $c_Q/k$ at $a = 1$), best-response effort saturates at the physical boundary $m^* = k$.

## 4.2 The first-best benchmark and physical feasibility

Before analyzing either strategic regime, we characterize the fully-informed, fully-aligned social optimum:

$$(m^{FB}(\kappa), a^{FB}(\kappa)) \;=\; \arg\max_{m\in[0,k],\, a\in[0,1]} U(m,a;\kappa).$$

**Proposition 3** (First-Best Policy on the Feasible Domain)**.** *On the physically feasible domain $m \in [0, k]$, under the linear-risk specification with $\Lambda > c_Q$:*

*(i) Asking weakly dominates guessing for every cost type $\kappa > 0$: $a^{FB}(\kappa) = 1$, with optimal specification $m^{FB}(\kappa) = \min\{k, c_Q/\kappa\}$.*

*(ii) For any type with $\kappa > c_Q/k$, asking strictly dominates guessing: $U(m^*(1), 1) > U(m^*(0), 0)$. For boundary types with $\kappa \leq c_Q/k$, specification effort saturates at $m^{FB} = k$, leaving zero residual attributes ($k - m = 0$) and rendering the asking policy payoff-irrelevant.*

*Proof.* For any fixed specification effort $m \in [0, k]$, the partial derivative of utility with respect to $a$ is:

$$\frac{\partial U}{\partial a} \;=\; (\Lambda - c_Q)(k - m).$$

Because $\Lambda > c_Q$, this derivative is strictly positive for all $m < k$ and zero at $m = k$. Hence, for every $m \in [0, k]$, $a = 1$ maximizes $U(m, \cdot\,; \kappa)$ over $[0, 1]$, so $a^{FB} = 1$ and

$$m^{FB} \;=\; \arg\max_{m \in [0,k]} U(m, 1; \kappa) \;=\; \min\{k, c_Q/\kappa\},$$

proving part (i). To quantify the gain over pure guessing (part (ii)), compare the optimized payoffs under $a \in \{0, 1\}$. Under $a = 1$, the user chooses $m^*(1) = \min\{k, c_Q/\kappa\}$. Under $a = 0$, the user chooses $m^*(0) = \min\{k, \Lambda/\kappa\}$. We evaluate the three type intervals:

- If $\kappa \geq \Lambda/k$, both best responses lie in $[0, k]$: $m^*(1) = c_Q/\kappa < m^*(0) = \Lambda/\kappa \leq k$. Evaluating the welfare difference:

$$U(m^*(1), 1; \kappa) - U(m^*(0), 0; \kappa) \;=\; (\Lambda - c_Q)\left(k - \frac{\Lambda + c_Q}{2\kappa}\right).$$

  Since $\kappa \geq \Lambda/k > \frac{\Lambda + c_Q}{2k}$, the bracketed term satisfies $k - \frac{\Lambda + c_Q}{2\kappa} > k - k = 0$. Hence $U(m^*(1), 1; \kappa) > U(m^*(0), 0; \kappa)$ strictly.

- If $c_Q/k < \kappa < \Lambda/k$, unconstrained effort under $a = 0$ exceeds $k$, so specification under guessing saturates at the boundary $m^*(0) = k$, yielding $U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$. Under asking, effort remains interior at $m^*(1) = c_Q/\kappa < k$, yielding $U(m^*(1), 1; \kappa) = V - c_Q k + \frac{c_Q^2}{2\kappa}$. Subtracting:

$$U(m^*(1), 1; \kappa) - U(k, 0; \kappa) \;=\; \frac{\kappa k^2}{2} - c_Q k + \frac{c_Q^2}{2\kappa} \;=\; \frac{(\kappa k - c_Q)^2}{2\kappa} \;>\; 0.$$

- If $\kappa \leq c_Q/k$, specification effort saturates at $m^* = k$ under both policies, so $U(k, 1; \kappa) = U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$.

Therefore, $a^{FB}(\kappa) = 1$ is globally optimal across all types $\kappa > 0$, with indifference only on the boundary set $\kappa \leq c_Q/k$ where $m^{FB} = k$. □

**Remark 4** (The Unclipped Threshold $\kappa^*$ and Genuine Asking–Guessing Trade-offs)**.** *In unconstrained algebra without the feasibility constraint $m \leq k$, setting the difference $(\Lambda - c_Q)(k - \frac{\Lambda + c_Q}{2\kappa})$ to zero yields the formal threshold $\kappa^* \equiv \frac{\Lambda + c_Q}{2k}$. However, since $\Lambda > c_Q$, we have $\kappa^* < \Lambda/k$ strictly. The region $\kappa < \kappa^*$ where guessing formally appeared to beat asking in unclipped algebra lies entirely in the unphysical domain where unconstrained effort $\Lambda/\kappa$ exceeds $k$ (the user would specify more attributes than exist in the task). Once $m \leq k$ is enforced, asking weakly dominates guessing*

*everywhere under linear risk because asking resolves uncertainty at lower unit cost ($c_Q < \Lambda$) while allowing the user to economize on specification effort.*

*Genuine interior thresholds, at which $a^{FB}$ switches between 0 and 1 across the type space, arise in two natural richer specifications:*

1. Exact conjunctive success: *Under $Vq(a,g)^{k-m} - c(m) - ac_Q(k-m)$, the marginal gain from clarification is $(k-m)[V(1-g)q^{k-m-1} - c_Q]$. When baseline accuracy $g$ is high or $k-m$ is small, this marginal gain turns negative, making guessing strictly optimal for low-cost types who specify most attributes upstream.*

2. Heterogeneous attribute stakes: *If attributes vary in importance with stakes $\Lambda_i$, efficiency prescribes asking only on high-stakes attributes ($\Lambda_i > c_Q$) and guessing on low-stakes attributes ($\Lambda_i < c_Q$).*

# 5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0,1]$ and cannot condition on user-specific attributes or observed effort. This provides a natural benchmark for policies embedded uniformly at training time or applied without user-level profiling.

## 5.1 User best response

The FOC $\partial U/\partial m = \Lambda(1-a) - \kappa m + ac_Q = 0$ gives, with $\beta(\kappa) = \Lambda/\kappa$ and $\gamma(\kappa) = (\Lambda - c_Q)/\kappa$,

$$m^*(\kappa; a) \;=\; \beta(\kappa) - a\,\gamma(\kappa), \qquad \frac{\partial m^*}{\partial a} \;=\; -\gamma(\kappa) \;=\; \frac{c_Q - \Lambda}{\kappa}.$$

(Here $\beta(\kappa)$ and $\gamma(\kappa)$ denote the intercept and slope of the best response; the two-argument $\beta(\kappa; a)$ used in the proof of Proposition 2 equals $\beta(\kappa) - a\gamma(\kappa)$.) The closed forms below assume interior specification levels $m^* \in (0,k)$ for every type and every $a \in [0,1]$, which holds when the support satisfies $\kappa_L \geq \Lambda/k$; for lower-cost types the best response saturates at $m^* = k$. The population moments $\bar{R}_0$ and $\bar{\gamma}$ are then exact. When clipping binds they are not, and we re-solve the leader's problem numerically on the clipped best responses wherever a result is stated for such a calibration (Remark 6, Section 7.1). The corner property of the unbiased case survives clipping exactly (Corollary 4).

**Remark 5** (Sign of the ask-rate effect on specification)**.** *Unlike the effect of guessing accuracy in Corollary 1, the sign of $\partial m^*/\partial a$ here does not depend on type or on $m$: it is positive for* every *type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that condition holds, a higher population ask rate induces users to specify* more*, not less; they pre-empt costly clarifying questions by articulating the answer directly. This introduces a new economic force relative to Section 2, where $a$ operated solely through effective accuracy, and generates a distinct, directly testable prediction (prediction P1′ below).*

## 5.2 The leader's problem

Let $R(a,\kappa) = k - m^*(\kappa; a)$ denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate $a \in [0,1]$ applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.1 at the follower's optimal best response $m^*(\kappa; a) = \beta(\kappa) - a\,\gamma(\kappa)$:

$$\Pi_\kappa(m^*(\kappa;a),a) \;=\; \mu_A\,U(m^*(\kappa;a),a;\kappa) \;-\; (\lambda_A - c_Q)\,a\,\big(k - m^*(\kappa;a)\big).$$

Substituting $U(m^*(\kappa;a),a;\kappa) = V - \Lambda(1-a)(k-m^*) - \frac{\kappa}{2}(m^*)^2 - a\,c_Q\,(k-m^*)$, using $k - m^*(\kappa;a) = R_0(\kappa) + a\,\gamma(\kappa)$, and defining the effective resolution stake $s \equiv \Lambda - c_Q$ and the assistant's excess asking friction $b \equiv \lambda_A - c_Q$, expanding in powers of $a$ yields:

$$\Pi_\kappa(m^*(\kappa;a),a) \;=\; \mu_A\left(V - \Lambda k + \frac{\Lambda^2}{2\kappa}\right) \;+\; (\mu_A s - b)\,R_0(\kappa)\,a \;+\; \frac{(\mu_A s - 2b)\,\gamma(\kappa)}{2}\,a^2.$$

Averaging over $\kappa \sim F$ requires no moments beyond the population sufficient statistics already defined, $\bar{R}_0 = k - \mathbb{E}[\beta]$ and $\bar{\gamma} = \mathbb{E}[\gamma] = s\,\mathbb{E}[1/\kappa]$:

$$\Pi(a) \;=\; \text{const} \;+\; \alpha_1\,a \;+\; \alpha_2\,a^2,$$

where $\text{const} = \mu_A(V - \Lambda k) + \frac{\mu_A \Lambda^2}{2}\mathbb{E}[1/\kappa]$, and

$$\alpha_1 \;\equiv\; (\mu_A s - b)\,\bar{R}_0, \qquad \alpha_2 \;\equiv\; \frac{(\mu_A s - 2b)\,\bar{\gamma}}{2}.$$

**Lemma 1** (Leader Objective as User Welfare with an Asking Friction Wedge)**.** *For any allocation $(m, a)$, the leader's objective satisfies:*

$$\Pi(m,a) \;=\; \mu_A\Big[U(m,a) \;-\; \tilde{\rho}\,a\,(k-m)\Big],$$

*where $\tilde{\rho} \equiv \frac{\lambda_A - c_Q}{\mu_A}$ is the normalized excess asking friction. The leader's objective incorporates true user welfare but introduces a per-question friction wedge $\tilde{\rho}$. In the leader's evaluation, clarifying questions are priced at an effective cost $c_Q^{eff} \equiv c_Q + \tilde{\rho}$. However, this is not an equivalence to a benevolent social planner, because followers continue to best-respond against their true private question cost $c_Q$, not $c_Q^{eff}$. Because $\mu_A > 0$ enters strictly as an overall positive scalar multiplier on the bracketed term, optimal leader policies depend on $(\mu_A, \lambda_A)$ solely through the excess friction ratio $\tilde{\rho}$.*

**Proposition 4** (Stackelberg-optimal pooling rate)**.** *Provided $\alpha_2 < 0$ (the second-order condition for strict concavity, equivalent to $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$), the leader's unconstrained stationary point is*

$$a^{SE} \;=\; -\frac{\alpha_1}{2\alpha_2} \;=\; -\frac{(\mu_A s - b)\,\bar{R}_0}{(\mu_A s - 2b)\,\bar{\gamma}}.$$

*When this stationary point lies in $[0,1]$, it is the unique global maximum of $\Pi(a)$ on $[0,1]$.*

**Corollary 2** (Friction bias suppresses the pooling ask rate on the interior branch)**.** *On the interior branch $\alpha_2 < 0$, differentiating the closed form in Proposition 4 directly with respect to $\lambda_A$ (holding primitives $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$ fixed) gives:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} \;=\; -\frac{\bar{R}_0}{\bar{\gamma}}\,\frac{\mu_A s}{(\mu_A s - 2b)^2} \;=\; -\frac{\bar{R}_0}{\bar{\gamma}}\,\frac{\mu_A(\Lambda - c_Q)}{\big[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)\big]^2}.$$

*For populations supported on the interior specification domain ($\kappa \geq \Lambda/k$ for all types in the support), follower effort satisfies $m^*(\kappa; 0) = \Lambda/\kappa \leq k$, ensuring that expected residual unspecified attributes*

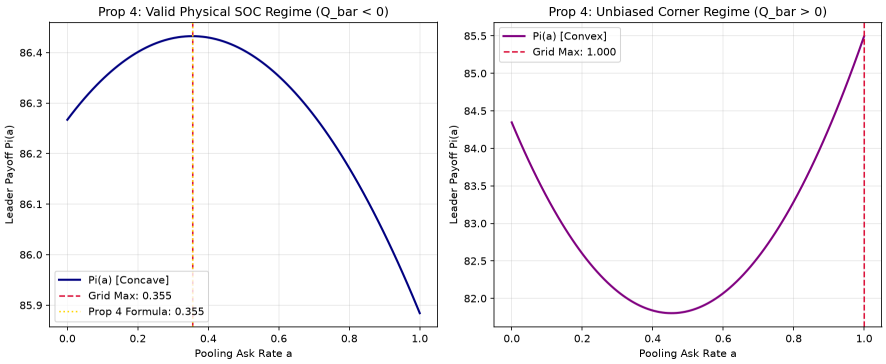



Figure 1: Leader pooling payoff $\Pi(a)$ and equilibrium ask rate $a^{SE}$ under Model I, comparing the unclipped closed form against numerical grid optimization.

*under silence satisfy* $\bar{R}_0 = k - \mathbb{E}[\beta] \geq 0$*, strictly unless every type sits exactly at* $\kappa = \Lambda/k$*. Because* $\bar{\gamma} > 0$*,* $\mu_A > 0$*, and* $s > 0$*, it follows that whenever* $\bar{R}_0 > 0$*:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} \; < \; 0.$$

*On this interior domain, higher perceived clarification friction* $\lambda_A$ *strictly suppresses the assistant's equilibrium asking rate. A clipping-robust weak version follows from a single-crossing argument:* $\partial \Pi_\kappa(m^*(\kappa; a), a)/\partial \lambda_A = -a(k - m^*(\kappa; a))$*, and* $a(k - m^*(\kappa; a))$ *is non-decreasing in* $a$ *(because* $m^*$ *is non-increasing in* $a$ *when* $s > 0$*, with or without clipping). Hence* $\Pi(a)$ *has decreasing differences in* $(a, \lambda_A)$*, and by Topkis's theorem the set of optimal pooling rates is weakly decreasing in* $\lambda_A$ *for any type distribution, including populations with clipped low-cost types.*

**Corollary 3** (Continuous transition of the pooling ask rate across friction regimes)**.** *Under asking friction bias (*$\lambda_A > c_Q$*), the second-order coefficient satisfies* $\alpha_2 < 0$ *if and only if* $\tilde{\rho} > s/2$*. For a continuum population supported on the interior specification domain (*$\kappa \geq \Lambda/k$*, so that* $\bar{R}_0 > 0$*), the endpoint crossover threshold satisfies* $\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}} > s/2$ *strictly. Consequently, on the entire convex branch (*$\tilde{\rho} \leq s/2 < \tilde{\rho}^*$*), the leader strictly prefers universal clarification (*$\Pi(1) > \Pi(0)$*), selecting* $a^{SE} = 1$ *unconditionally. On the strictly concave branch (*$\tilde{\rho} > s/2$*), the optimal pooling policy transitions continuously on* $[0, 1]$ *across three distinct friction regimes:*

1. Mild bias ($\tilde{\rho} \leq \tilde{\rho}_1$): *For* $\tilde{\rho} \leq \tilde{\rho}_1 \equiv \frac{s(\bar{R}_0 + \bar{\gamma})}{\bar{R}_0 + 2\bar{\gamma}}$*, the unconstrained stationary point satisfies* $-\alpha_1/(2\alpha_2) \geq 1$*, so the constrained maximum on* $[0, 1]$ *is universal clarification:* $a^{SE} = 1$*.*

2. Moderate bias ($\tilde{\rho}_1 < \tilde{\rho} < s$): *The unconstrained stationary point is strictly interior, so the optimal ask rate is uniquely:*

$$a^{SE} \; = \; -\frac{\alpha_1}{2\alpha_2} \; = \; \frac{(s - \tilde{\rho})\bar{R}_0}{(2\tilde{\rho} - s)\bar{\gamma}} \; \in \; (0, 1),$$

*which is strictly decreasing in friction (*$\partial a^{SE}/\partial \tilde{\rho} < 0$*). At the threshold value* $\tilde{\rho} = \tilde{\rho}^* \in (\tilde{\rho}_1, s)$*, endpoint payoffs satisfy* $\Pi(1) = \Pi(0)$*, which corresponds geometrically to the quadratic vertex lying exactly at* $a^{SE} = 1/2$*. The interior optimum transitions smoothly through this point without any regime discontinuity.*

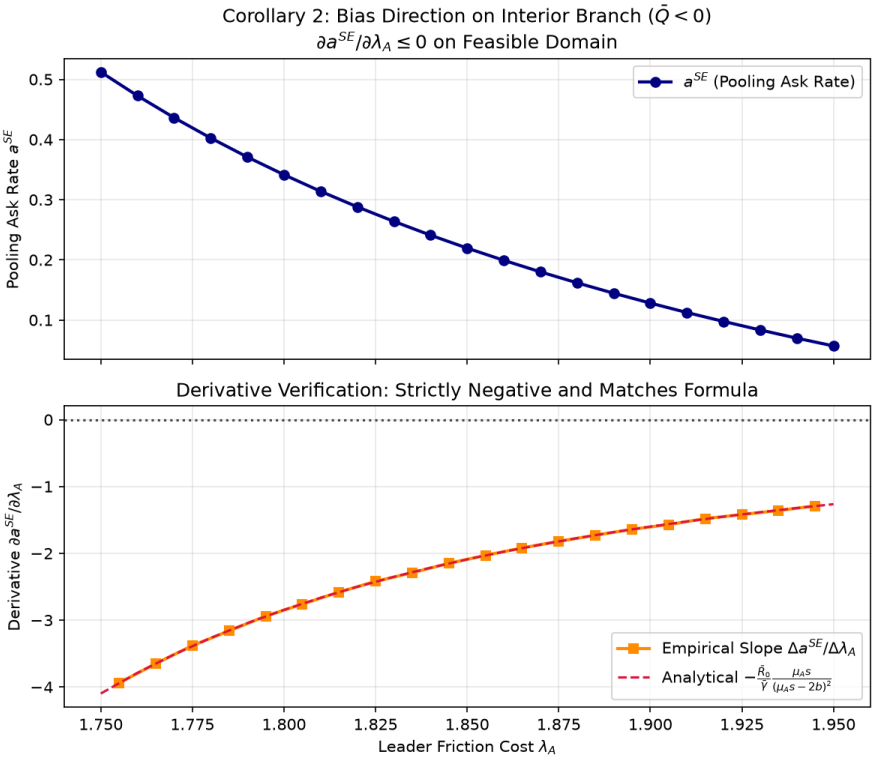



Figure 2: Perceived asking friction $\lambda_A$ suppresses the Stackelberg pooling ask rate ($\partial a^{SE}/\partial \lambda_A \leq 0$) across varying altruism levels $\mu_A$.

*3.* Prohibitive friction ($\tilde{\rho} \geq s$, equivalently $c_Q^{eff} \geq \Lambda$): *The leader unconstrainedly prefers zero clarification ($-\alpha_1/(2\alpha_2) \leq 0$), shutting down asking completely: $a^{SE} = 0$.*

*This continuous transition characterizes the single unclipped quadratic representation with continuous type support; in discrete type distributions (such as two-type populations $\kappa \in \{\kappa_L, \kappa_H\}$) or under boundary clipping, piecewise best-response regimes can induce discrete policy shifts.*

**Corollary 4** (Pooling is a corner solution in the unbiased case)**.** *In the unbiased case ($\mu_A = 1, \lambda_A = c_Q \implies b = 0$), the quadratic coefficient is:*

$$\alpha_2 \;=\; \frac{s\,\bar{\gamma}}{2} \;=\; \frac{s^2\,\mathbb{E}[1/\kappa]}{2} \;=\; \frac{(\Lambda - c_Q)^2\,\mathbb{E}[1/\kappa]}{2} \;\geq\; 0 \quad \textit{always.}$$

*The second-order condition for an interior maximum therefore fails: $\Pi(a)$ is weakly convex on $[0,1]$, its stationary point is a local minimum, and the true optimum is always a corner, $a^{SE} \in \{0,1\}$, not an interior compromise. Comparing the two corners directly in unclipped algebra:*

$$\Pi(1) - \Pi(0) \;=\; \alpha_1 + \alpha_2 \;=\; s\left(\bar{R}_0 + \frac{\bar{\gamma}}{2}\right) \;=\; (\Lambda - c_Q)\left[k - \frac{\Lambda + c_Q}{2}\,\mathbb{E}\left[\frac{1}{\kappa}\right]\right],$$

*which yields the formal unclipped threshold $k^* \equiv \frac{\Lambda + c_Q}{2}\mathbb{E}[1/\kappa]$. When follower best responses are physically constrained to the feasible domain $m \leq k$, Proposition 3 implies that asking ($a = 1$) weakly dominates guessing for every individual type in the support, so unbiased pooling under physical feasibility selects $a^{SE} = 1$ unconditionally. Genuine interior trade-offs between asking and guessing at the population level emerge under the exact conjunctive success model, as analyzed in Section 10.*

*The corner property itself does not rely on the unclipped algebra. For each $m$, $U(m, a; \kappa)$ is affine in $a$, so the clipped value $U^*(a; \kappa) = \max_{m \in [0,k]} U(m, a; \kappa)$ is a pointwise maximum of affine functions and hence convex in $a$. With $\Pi = \mathbb{E}_\kappa[U^*(a; \kappa)]$ in the unbiased case, $\Pi$ is convex on $[0,1]$ for* any *type distribution, with or without boundary clipping, and its maximum is attained at a corner.*

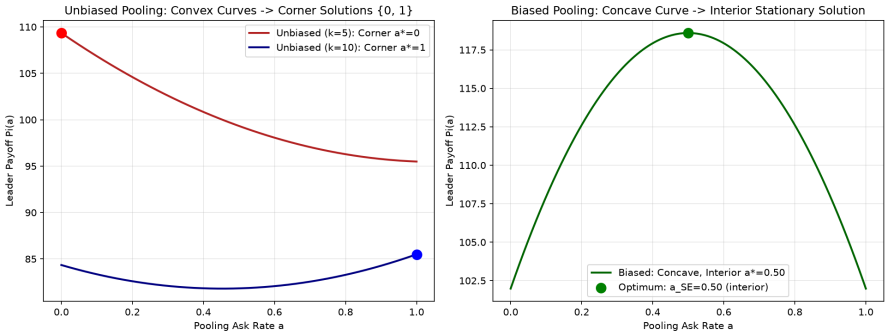



Figure 3: Payoff curvature across friction regimes: weakly convex corner solution in the unbiased regime ($\mu_A = 1, \lambda_A = c_Q$) versus strictly concave interior optimum under friction bias ($\lambda_A > c_Q$).

**Proposition 5** (Equilibrium Reversal of Guessing Capability: The Guessing Trap)**.** *Under asking friction bias ($\lambda_A > c_Q$), on the concave interior branch ($\alpha_2 < 0$, $a^{SE} \in (0,1)$) with every type's best*

*response interior ($m^*(\kappa) \in (0,k)$), an increase in assistant guessing capability $g$ affects the pooling clarification rate according to:*

$$\frac{da^{SE}}{dg} \;<\; 0 \iff \mu_A \bar{R}_0(g) \;>\; (\mu_A s(g) - b)\,\mathbb{E}\left[\frac{1}{\kappa}\right]\left(1 - 2a^{SE}\right).$$

*In particular:*

*(i) On the interior branch with $\bar{R}_0 > 0$, if $a^{SE} \geq 1/2$ the condition holds automatically, so $\frac{da^{SE}}{dg} < 0$. If $a^{SE} < 1/2$, the derivative can reverse when $\bar{R}_0$ is small.*

*(ii) When $\frac{da^{SE}}{dg} < 0$, prompt lengthening and welfare decline are mathematically equivalent and hold simultaneously if and only if*

$$-\frac{da^{SE}}{dg} \;>\; \frac{L(1 - a^{SE})}{\Lambda(g) - c_Q},$$

*reversing the comparative statics of Proposition 1 in Stackelberg equilibrium:*

- ***Equilibrium prompt lengthening:*** $\frac{d\mathbb{E}[m^*]}{dg} > 0$.
- ***Equilibrium welfare decline:*** $\frac{dW}{dg} < 0$.

*Proof.* On the interior branch, the stationary point is $a^{SE}(g) = -\frac{\alpha_1(g)}{2\alpha_2(g)} = \frac{(\mu_A s(g) - b)\bar{R}_0(g)}{(2b - \mu_A s(g))\bar{\gamma}(g)}$, where $\Lambda(g) = L(1-g)$, $s(g) = L(1-g) - c_Q$, $\bar{R}_0(g) = k - \Lambda(g)\mathbb{E}[1/\kappa]$, and $\bar{\gamma}(g) = s(g)\mathbb{E}[1/\kappa]$. Differentiating with respect to $g$ and collecting terms in $\mu_A, b, s(g), \bar{R}_0(g)$, and $\mathbb{E}[1/\kappa]$ yields:

$$\frac{da^{SE}}{dg} \;=\; -\frac{L}{\bar{\gamma}(g)(2b - \mu_A s(g))}\left[\mu_A \bar{R}_0(g) \;-\; (\mu_A s(g) - b)\mathbb{E}\left[\frac{1}{\kappa}\right]\left(1 - 2a^{SE}\right)\right].$$

Because the prefactor $-\frac{L}{\bar{\gamma}(g)(2b - \mu_A s(g))}$ is strictly negative on the concave interior branch ($2b > \mu_A s$), it follows that $da^{SE}/dg < 0$ if and only if $\mu_A \bar{R}_0(g) > (\mu_A s(g) - b)\mathbb{E}[1/\kappa](1 - 2a^{SE})$. When $a^{SE} \geq 1/2$, $(1 - 2a^{SE}) \leq 0$. Interiority with $\bar{R}_0 > 0$ and $2b > \mu_A s$ requires $\mu_A s > b$ (otherwise $a^{SE} \leq 0$), so the right-hand side is non-positive while $\mu_A \bar{R}_0 > 0$, and the condition holds.

By the chain rule, total equilibrium follower effort satisfies:

$$\frac{dm^*}{dg} \;=\; \frac{\partial m^*}{\partial g} \;+\; \frac{\partial m^*}{\partial a}\frac{da^{SE}}{dg} \;=\; \frac{1}{\kappa}\left[-L(1 - a^{SE}) \;-\; s(g)\frac{da^{SE}}{dg}\right].$$

Similarly, by the envelope theorem on follower choice of $m^*$, total equilibrium user welfare $W(g) \equiv \mathbb{E}[U(m^*(g), a^{SE}(g); g)]$ satisfies:

$$\frac{dW}{dg} \;=\; \mathbb{E}\left[\left.\frac{\partial U}{\partial g}\right|_{m^*}\right] \;+\; \mathbb{E}\left[\left.\frac{\partial U}{\partial a}\right|_{m^*}\right]\frac{da^{SE}}{dg} \;=\; \mathbb{E}[(k - m^*)]\left[L(1 - a^{SE}) \;+\; s(g)\frac{da^{SE}}{dg}\right].$$

Because $\mathbb{E}[k - m^*] > 0$ and $\mathbb{E}[1/\kappa] > 0$, prompt lengthening ($d\mathbb{E}[m^*]/dg > 0$) and welfare decline ($dW/dg < 0$) are mathematically equivalent and hold simultaneously if and only if $-da^{SE}/dg > \frac{L(1-a^{SE})}{s(g)}$. □

**Remark 6** (Analytical Conditions versus Clipped Numerical Calibration)**.** *Proposition 5 characterizes the unclipped interior branch where best responses remain interior ($m^*(\kappa) < k$). When $a^{SE} < 1/2$, the sign condition can fail, so it is not vacuous. For example, with many-valued attributes where guessing accuracy satisfies $g \in [0.05, 0.10]$ (with $k = 12, L = 17, c_Q = 4.5, \lambda_A = 11.5$, $\mu_A = 1$, and $\kappa \sim$ Uniform$[1.4, 1.5]$), all types remain unclipped and $a^{SE}$ rises from 0.211 to 0.230 as $g$ increases.*

*In the benchmark calibration ($k = 10, L = 10, c_Q = 2, \mu_A = 1, b = 1.2, \kappa \sim$ Uniform$[0.2, 0.6]$), unclipped moments at $g = 0.5$ give $\bar{R}_0 = -3.73$ and $\alpha_2 = +2.47$, so the closed form of Proposition 4 does not apply. On the binary-attribute range $g \in [0.50, 0.65]$, low-cost followers are subject to boundary clipping $m^*(\kappa) = \min\{k, \dots\}$, and we solve the leader's problem directly on the clipped best responses. Evaluating the exact clipped model via numerical quadrature (`underspec_sim/verifications/verify_prop5_guessing_trap.py`): as guessing capability improves from $g = 0.50$ to $g = 0.60$, the equilibrium ask rate collapses from $a^{SE} = 0.661$ to $a^{SE} = 0.424$. In response to this withdrawal of clarification, average prompt specification expands from $\mathbb{E}[m] = 7.728$ to 7.953, and expected user welfare declines from $W = 82.134$ to 81.843. To gauge how common the trap is, we draw 2,000 random parameter vectors (sampling ranges in `verify_guessing_trap_incidence.py`; $k \in \{6, \dots, 15\}$, $g \in [0.5, 0.8]$, $\tilde{\rho} \in [0, 1.5\,s]$, uniform type distributions) and re-solve the clipped pooling equilibrium at $g$ and $g + 0.02$. Simultaneous prompt lengthening and welfare reduction occur in 12.2% of all draws and in 51.6% of the 318 draws with an interior $a^{SE}$. These frequencies depend on the sampling distribution and should be read as evidence that the trap is not a knife-edge, not as population estimates.*

This structure matters for the comparison in Section 7: under linear risk, Model I falls short of the first best only through objective bias, which lowers $a^{SE}$ (Corollary 2) and suppresses the operational clarification rate.

# 6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types $\kappa_L < \kappa_H$ with population shares $f_L, f_H$ (the continuum-type extension using the standard Mirrlees technique is discussed in Section 10). Let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa: \;\; U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \qquad \text{IR}_\kappa: \;\; U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$. We normalize the follower's reservation utility to $\underline{U} = 0$, representing complete task abandonment without AI assistance (with $V$ large enough that this constraint never binds). The economically relevant outside option is stronger. Because users can type any prompt, the assistant must also specify a response to off-menu specification levels, and the harshest admissible one is not to ask ($a = 0$). A user can therefore always secure $\underline{U}(\kappa) \equiv \max_{m \in [0,k]} U(m, 0; \kappa)$, which equals $V - \frac{\kappa}{2}k^2$ for $\kappa \leq \Lambda/k$ and $V - \Lambda k + \frac{\Lambda^2}{2\kappa}$ otherwise. Lemma 2 below shows that the optimal menu respects this endogenous outside option for both types, so it can be implemented with free prompt choice.

## 6.1 Single-crossing

The marginal rate of substitution between $m$ and $a$ along an indifference curve in $(m, a)$-space is given by the slope $\text{MRS}(m, a; \kappa) \equiv -\frac{\partial U/\partial m}{\partial U/\partial a}$. From user utility $U(m, a; \kappa) = V - \Lambda(1 - a)(k -$

$m) - \frac{\kappa}{2}m^2 - ac_Q(k-m)$, we have $\partial U/\partial m = \Lambda(1-a) + ac_Q - \kappa m$ and $\partial U/\partial a = s(k-m)$, where $s \equiv \Lambda - c_Q > 0$ is the net social gain from asking. Hence,

$$\mathrm{MRS}(m, a; \kappa) \;=\; \frac{\kappa m - [\Lambda - as]}{s(k-m)}, \qquad \frac{\partial \mathrm{MRS}}{\partial \kappa} \;=\; \frac{m}{s(k-m)} \;>\; 0,$$

which holds strictly for all interior specification levels $0 < m < k$ under the maintained condition $s > 0$. Higher-$\kappa$ types therefore have strictly steeper indifference curves in $(m, a)$-space: they require more compensating clarification service $a$ per unit increase in specification effort $m$. This satisfies the standard Spence–Mirrlees single-crossing property (Spence, 1973) and underwrites the screening analysis below.

### 6.2 No distortion without bias

**Proposition 6** (Efficiency survives private information when the leader is unbiased)**.** *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot;\kappa)$ for every $\kappa$), then implementing each type's own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

*Proof.* $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot;\kappa)$ by definition, so for any $\kappa' \neq \kappa$,

$$U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \;\geq\; U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$$

holds trivially: no bundle can give type $\kappa$ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot;\kappa)$, the leader's payoff from serving each true type is maximized at that type's own first best, so this menu is also leader-optimal. □

This result highlights a fundamental distinction between human–AI coordination and classical adverse selection (such as Baron–Myerson or Laffont–Tirole). In regulatory and auction models, the principal pays monetary transfers to extract surplus and thus treats informational rents as a pure loss. Here, by contrast, the leader's objective can coincide directly with user welfare; when it does, incentive compatibility is costless, and private information incurs no welfare penalty. Distortion in this setting is therefore a pure signature of objective misalignment, not of asymmetric information per se.

**Remark 7** (Degeneracy of the Unbiased Menu on the Asking Margin)**.** *Under Proposition 3, $a^{FB}(\kappa) = 1$ for every type in the population on the physically feasible domain $m \leq k$. The unbiased first-best menu is therefore $(m^{FB}(\kappa), 1) = (\min\{k, c_Q/\kappa\}, 1)$, which offers uniform asking $a = 1$ across all types and leaves followers to freely choose their preferred specification effort. This menu does not screen across the asking margin: scalar pooling at $a^{SE} = 1$ (Corollary 4) implements each type's first best identically. Consequently, under linear risk on the feasible domain, the welfare gap between pooling and screening at zero bias is identically zero ($\Pi_{II} - \Pi_I = 0$). A positive instrument-restriction gap at zero bias emerges only in settings where first best itself prescribes different asking policies across types, such as the exact conjunctive success model (Section 10) or under attribute loss heterogeneity.*

### 6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot;\kappa)$ (bias present), the leader's preferred bundle per type,

$$(m^B_\kappa, a^B_\kappa) \;=\; \arg\max_{m,a} \Pi_\kappa(m, a)$$

(computable in closed form by the same method as Proposition 3, with $\Lambda, c_Q$ replaced by their $\mu_A, \lambda_A$-weighted counterparts), no longer automatically satisfies IC, because $\Pi_\kappa$ and the user's true $U(\cdot;\kappa)$ now rank bundles differently.

**Proposition 7** (Second-Best Screening Menu and Distortion Regimes)**.** *Assume $\Lambda > c_Q$, under-asking bias ($0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$, so that $c_Q < c_Q^{eff} < \Lambda$), and cost heterogeneity $\kappa_L < c_Q^{eff}/k < \kappa_H$. Let population shares be $f_L, f_H > 0$ with $f_L + f_H = 1$, define the relative share weight $t \equiv \frac{f_H}{f_L}\frac{s-\tilde{\rho}}{s}$, and define the stationary candidate:*

$$m_H^* \;\equiv\; \frac{\Lambda\tilde{\rho}}{s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L}.$$

*Define the leader's unconstrained bundle for type $H$, $m_H^B \equiv c_Q^{eff}/\kappa_H$ (with $a_H^B = 1$), the complexity thresholds $k_{slack} \equiv \frac{2c_Q}{\kappa_L} - m_H^B$ and $k_{crit} \equiv \frac{2c_Q}{\kappa_L} - m_H^*$ (so $k_{slack} < k_{crit}$ if and only if $m_H^* < m_H^B$), and the saturation threshold $\bar{\kappa}_L \equiv \frac{c_Q^{eff} + tc_Q}{k(1+t)}$. Then the four regimes below partition the parameter space; in regimes 2–4, the closed forms describe the optimal menu whenever the saturation condition stated for that regime holds (otherwise item 5 applies):*

1. ***Unconstrained slack regime ($k \le k_{slack}$):*** *The unconstrained menu $((k,0),(m_H^B,1))$ satisfies $IC_L$ ($\kappa_L(k + m_H^B) \le 2c_Q$). Both types receive their unconstrained preferred bundles, with zero distortion in effort and asking. This regime contains all $\kappa_L \le c_Q/k$.*

2. ***Effort-distortion regime ($k_{slack} < k \le k_{crit}$, non-empty only when $m_H^* < m_H^B$):*** *Under the saturation condition $f_H\,(c_Q^{eff} - \kappa_H m_H^{SB}) \le f_L\,(c_Q^{eff} - \kappa_L k)$, $IC_L$ binds, type $L$ receives $(k,0)$, and asking remains at the leader-preferred level $a_H = 1$. The distortion is purely a downward effort distortion on the high-cost type:*

$$m_H^{SB} \;=\; \frac{2c_Q}{\kappa_L} - k \;<\; m_H^B, \qquad a_H^{SB} \;=\; 1.$$

3. ***Asking-distortion regime ($k > \max\{k_{slack}, k_{crit}\}$ and $k \ge m_H^*$):*** *Under the saturation condition $\kappa_L \le \bar{\kappa}_L$, the leader sets $(m_L^{SB}, a_L^{SB}) = (k,0)$, and the high-cost contract is unique and given in closed form by:*

$$m_H^{SB} \;=\; m_H^*, \qquad a_H^{SB} \;=\; 1 \;-\; \frac{\frac{\kappa_L(k+m_H^*)}{2} - c_Q}{s} \;\in\; \Big(1 - \frac{\tilde{\rho}}{s},\, 1\Big).$$

   *The asking distortion is therefore bounded by the relative bias: $0 < 1 - a_H^{SB} < \tilde{\rho}/s$. It intensifies as $\kappa_L$ increases toward $\bar{\kappa}_L$. At $m_L = k$, zero residual attributes remain ($k - m_L = 0$), so $a_L$ is payoff-irrelevant to both parties.*

4. ***Full-specification regime ($k_{slack} < k < m_H^*$):*** *When cost heterogeneity is narrow, $m_H^* > k$ and the feasibility constraint $m_H \le k$ binds. Under the saturation condition $\kappa_L \le \bar{\kappa}_L$, both types specify fully, $m_L^{SB} = m_H^{SB} = k$; residual attributes vanish, asking is payoff-irrelevant, and the menu is payoff-equivalent to pooling at $(k,0)$.*

5. ***Boundary relaxation:*** *When the saturation condition of regime 2, 3, or 4 fails (in regimes 3–4, when $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the leader instead offers type $L$ clarification service ($a_L^{SB} = 1$, $m_L^{SB} < k$). This raises $U_L$, relaxes $IC_L$, and weakly moderates the distortion on type $H$ relative to the saturation formula without reversing its sign.*

*In regimes 1–4, type $L$ is held exactly to its outside option, $U_L = \underline{U}(\kappa_L)$, while type $H$ earns a strictly positive rent over $\underline{U}(\kappa_H)$ in regimes 1–3 (except at the knife-edge $k = m_H^*$).*

**Lemma 2** (Implementability with free prompt choice)**.** *Suppose the assistant answers every off-menu specification level with $a = 0$. Then, under the hypotheses of Proposition 7, no type gains by choosing an off-menu prompt: each type's menu bundle yields at least $\underline{U}(\kappa) = \max_{m\in[0,k]} U(m, 0; \kappa)$.*

*Proof.* For type $L$, $\kappa_L k < c_Q^{eff} < \Lambda$, so $U(\cdot, 0; \kappa_L)$ is increasing on $[0, k]$ and $\underline{U}(\kappa_L) = U(k, 0; \kappa_L)$; type $L$'s bundle yields at least this (with equality at $(k, 0)$, and strictly more under boundary relaxation). For type $H$, let $m^o \equiv \min\{k, \Lambda/\kappa_H\}$ be its unassisted optimum. In every regime the assigned effort satisfies $m_H^{SB} \le m^o$: $m_H^B = c_Q^{eff}/\kappa_H < \Lambda/\kappa_H$, the effort-regime level lies below $m_H^B$, and $m_H^* < \Lambda/\kappa_H$ because $(s - \tilde\rho)(\kappa_H - \kappa_L) > 0$ (in regime 4, $k < m_H^* < \Lambda/\kappa_H$, so $m^o = k = m_H^{SB}$).

*Case 1: $IC_L$ binds* (regimes 2–5). Then $U(m_H^{SB}, a_H^{SB}; \kappa_L) = U_L \ge \underline{U}(\kappa_L) \ge U(m^o, 0; \kappa_L)$. Using $U(m, a; \kappa_H) = U(m, a; \kappa_L) - \frac{\kappa_H - \kappa_L}{2} m^2$,

$$U(m_H^{SB}, a_H^{SB}; \kappa_H) - U(m^o, 0; \kappa_H) \;\ge\; \frac{\kappa_H - \kappa_L}{2}\Big[(m^o)^2 - (m_H^{SB})^2\Big] \;\ge\; 0.$$

*Case 2: slack regime*, bundle $(m_H^B, 1)$. If $\kappa_H \le \Lambda/k$, then $m^o = k$ and $U(m_H^B, 1; \kappa_H) - U(k, 0; \kappa_H) = (k - m_H^B)\big[\frac{\kappa_H k + c_Q^{eff}}{2} - c_Q\big] > 0$. If $\kappa_H > \Lambda/k$, then $m^o = \Lambda/\kappa_H$, and direct substitution gives

$$U(m_H^B, 1; \kappa_H) - U(m^o, 0; \kappa_H) \;=\; sk - \frac{s(\Lambda + c_Q) + \tilde\rho^{\,2}}{2\kappa_H} \;>\; sk - \frac{2s\Lambda}{2\kappa_H} \;=\; s\Big(k - \frac{\Lambda}{\kappa_H}\Big) \;>\; 0,$$

using $\tilde\rho < s$. □

*Proof of Proposition 7.* We establish the result in five analytical steps:

**Step 1: Unconstrained biased optima.** The leader objective per type is $\Pi_\kappa(m, a) = \mu_A[V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - ac_Q^{eff}(k - m)]$, where $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A$. Under the maintained condition $0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$, effective asking friction satisfies $c_Q < c_Q^{eff} < \Lambda$. For type $H$ with $\kappa_H > c_Q^{eff}/k$, because $c_Q^{eff} < \Lambda$, $\Pi_{\kappa_H}$ is strictly increasing in $a$ for any $m < k$ (slope $\mu_A(\Lambda - c_Q^{eff})(k - m) > 0$). The leader's unconstrained optimum for type $H$ therefore sets $a_H^B = 1$ and interior specification $m_H^B = c_Q^{eff}/\kappa_H < k$. For type $L$ with $\kappa_L < c_Q^{eff}/k$, the partial derivative with respect to effort satisfies $\partial\Pi_{\kappa_L}/\partial m = \mu_A[\Lambda(1 - a) + ac_Q^{eff} - \kappa_L m] \ge \mu_A[c_Q^{eff} - \kappa_L k] > 0$ for all $m \in [0, k]$ and all $a \in [0, 1]$. Hence $\Pi_{\kappa_L}$ is strictly increasing in $m$, and attains its unconstrained maximum on $[0, k] \times [0, 1]$ at $m_L^B = k$. At $m_L^B = k$, zero attributes remain unspecified ($k - m_L^B = 0$), so asking is payoff-irrelevant to both the user and the leader ($\Pi_{\kappa_L}(k, a) = \mu_A[V - \frac{\kappa_L}{2}k^2]$ for all $a$). Setting $a_L^B = 0$ is the unconstrained benchmark choice because it gives zero asking service, minimizing the temptation of type $H$ to mimic type $L$.

**Step 2: Upward temptation to mimic ($IC_L$ violation).** Evaluating type $\kappa_L$'s payoff from mimicking the high type's unconstrained bundle $(m_H^B, 1)$ versus accepting its own bundle $(k, 0)$:

$$U(m_H^B, 1; \kappa_L) - U(k, 0; \kappa_L) \;=\; (k - m_H^B)\left[\frac{\kappa_L(k + m_H^B)}{2} - c_Q\right].$$

Because $m_H^B < k$, this difference is strictly positive if and only if $\kappa_L(k + m_H^B) > 2c_Q$, which is equivalent to $k > k_{\text{slack}} \equiv \frac{2c_Q}{\kappa_L} - m_H^B = \frac{2c_Q}{\kappa_L} - \frac{c_Q^{eff}}{\kappa_H}$. When $k \le k_{\text{slack}}$, $\text{IC}_L$ is strictly satisfied at the unconstrained bundles, establishing claim 1.

**Step 3: Relaxed problem and slackness of dropped constraints ($\text{IC}_H$, $\text{IR}_H$).** We formulate the relaxed screening problem by maximizing the expected leader objective subject to $\text{IC}_L$ alone, dropping $\text{IC}_H$, $\text{IR}_L$, and $\text{IR}_H$. We then establish that all dropped constraints are strictly satisfied at the relaxed optimum:

1. *Slackness of $IC_H$ via single-crossing:* By the Spence–Mirrlees sorting condition $\frac{\partial^2 U}{\partial m\,\partial \kappa} = -m < 0$, the rent identity for any menu $((m_L, a_L), (m_H, a_H))$ is:

$$U(m_H, a_H; \kappa_H) - U(m_L, a_L; \kappa_H) \;=\; [U(m_H, a_H; \kappa_L) - U(m_L, a_L; \kappa_L)] \;+\; \frac{\kappa_H - \kappa_L}{2}\big(m_L^2 - m_H^2\big).$$

When $(m_L, a_L) = (k, 0)$ and $\text{IC}_L$ binds with equality ($U(m_H, a_H; \kappa_L) = U(k, 0; \kappa_L)$), this simplifies directly to:

$$U(m_H, a_H; \kappa_H) - U(k, 0; \kappa_H) \;=\; \frac{\kappa_H - \kappa_L}{2}\big(k^2 - m_H^2\big) \;>\; 0,$$

which is strictly positive for any $m_H < k$ since $\kappa_H > \kappa_L$. Hence $\text{IC}_H$ holds with strict inequality (strict slack).

2. *Participation:* With $\underline{U} = 0$ and $V$ large, both IR constraints are slack. Against the stronger endogenous outside option $\underline{U}(\kappa) = \max_m U(m, 0; \kappa)$, Lemma 2 shows that both types weakly prefer their bundles; $\text{IR}_L$ holds with equality whenever $(m_L, a_L) = (k, 0)$, and $\text{IR}_H$ holds strictly whenever $m_H^{SB} < m^o$.

**Step 4: Global optimization along the $\text{IC}_L$ boundary and regimes.** Because $\Pi_{\kappa_H}$ is strictly increasing in $a_H$ on $c_Q^{eff} < \Lambda$, the leader pushes $a_H$ as high as incentive compatibility permits, so $\text{IC}_L$ must bind at the relaxed optimum. Along the binding boundary $U(m_H, a_H; \kappa_L) = U_L^0 \equiv V - \frac{\kappa_L}{2}k^2$, utility is strictly linear in $a_H$, uniquely determining:

$$a_H(m_H) \;=\; 1 \;-\; \frac{U(m_H, 1; \kappa_L) - U_L^0}{s(k - m_H)} \;=\; 1 \;-\; \frac{\frac{\kappa_L(k+m_H)}{2} - c_Q}{s}.$$

Substituting this boundary into the leader's objective $\Pi_{\kappa_H}(m_H, a_H(m_H))$ yields a strictly concave quadratic in $m_H$:

$$\frac{d^2\Pi_{\kappa_H}}{dm_H^2} \;=\; -\,\frac{\mu_A\big[s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L\big]}{s} \;<\; 0,$$

which attains its unique unconstrained stationary point at:

$$m_H^* \;=\; \frac{b\Lambda}{\mu_A s(\kappa_H - \kappa_L) + b\kappa_L} \;=\; \frac{\Lambda\tilde{\rho}}{s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L}.$$

Now consider the interaction between the constraints $m_H \le k$ and $a_H \le 1$:

- If $k_{\text{slack}} < k \le k_{\text{crit}} \equiv \frac{2c_Q}{\kappa_L} - m_H^*$ (which is non-empty whenever $m_H^* < m_H^B$), then $\frac{\kappa_L(k+m_H^*)}{2} \le c_Q$. Evaluating $a_H(m_H^*)$ would yield $a_H \ge 1$. Since $a_H \le 1$, the upper bound $a_H = 1$ binds. At $a_H = 1$, $\text{IC}_L$ reads $(k - m_H)[\frac{\kappa_L(k+m_H)}{2} - c_Q] \le 0$, i.e., $m_H \le \hat{m} \equiv \frac{2c_Q}{\kappa_L} - k$, and $\hat{m} < m_H^B$ because $k > k_{\text{slack}}$. On $m_H \le \hat{m}$, $\Pi_{\kappa_H}(m_H, 1)$ is increasing (as $\hat{m} < m_H^B$); on $m_H > \hat{m}$, the leader is on the boundary curve $a_H(m_H) < 1$, along which $\Pi_{\kappa_H}$ is concave with peak $m_H^* \le \hat{m}$ and hence decreasing. The maximizer is therefore the kink $m_H = \hat{m}$ with $a_H = 1$. The distortion manifests entirely through prompt effort, establishing claim 2. When $m_H^* \ge m_H^B$, $k_{\text{slack}} \ge k_{\text{crit}}$ and this intermediate effort-distortion regime is empty.

- If $k > \max\{k_{\text{slack}}, k_{\text{crit}}\}$ and $k \geq m_H^*$, then $\frac{\kappa_L(k+m_H^*)}{2} > c_Q$, yielding interior asking distortion $a_H^{SB} = a_H(m_H^*) < 1$ with effort $m_H^{SB} = m_H^*$, establishing claim 3.
- If $k < m_H^*$ (which occurs when cost heterogeneity $\kappa_H - \kappa_L$ is small), the stationary point exceeds the physical attribute count ($m_H^* > k$). By strict concavity of $\Pi_{\kappa_H}(m_H)$ on $[0, k]$, the constrained maximizer is $m_H^{SB} = k$. At $m_H = k$, residual attributes vanish ($k - m_H = 0$), so asking is payoff-irrelevant and both types specify fully, establishing claim 4.

We now determine when $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is optimal. Consider departing from $m_L = k$ by offering type $L$ clarification service ($a_L = 1$) with $m_L < k$. Since $\kappa_L > c_Q/k$, reducing $m_L$ below $k$ increases type $L$'s utility at marginal rate $\frac{\partial U}{\partial m_L}\Big|_{m_L=k} = c_Q - \kappa_L k < 0$, yielding a utility increase of $\kappa_L k - c_Q > 0$. With population shares $f_L, f_H$, increasing type $L$'s utility by $+1$ relaxes $\text{IC}_L$, allowing $a_H$ to increase by $\frac{1}{s(k-m_H)}$, which increases the leader's payoff on type $H$ at rate $f_H \frac{\mu_A s - b}{s}(\kappa_L k - c_Q)$. Conversely, granting type $L$ clarification costs the leader $f_L \mu_A (c_Q^{eff} - \kappa_L k)$ per unit of reduction in $m_L$. Departing from $(k, 0)$ is strictly unprofitable if and only if the marginal gain on type $H$ is weakly less than the marginal cost on type $L$:

$$f_H \frac{\mu_A s - b}{s}(\kappa_L k - c_Q) \;\leq\; f_L \mu_A (c_Q^{eff} - \kappa_L k).$$

Dividing by $f_L \mu_A$ and substituting $\tilde{\rho} = b/\mu_A$ and $t \equiv \frac{f_H}{f_L}\frac{s-\tilde{\rho}}{s}$, this simplifies to:

$$t(\kappa_L k - c_Q) \;\leq\; c_Q^{eff} - \kappa_L k \iff \kappa_L \;\leq\; \bar{\kappa}_L \;\equiv\; \frac{c_Q^{eff} + t c_Q}{k(1+t)}.$$

When $f_L = f_H = 0.5$, $t = \frac{s-\tilde{\rho}}{s}$ and $\bar{\kappa}_L$ simplifies to $\frac{s c_Q^{eff} + (s-\tilde{\rho}) c_Q}{k(2s-\tilde{\rho})}$. This local condition is also global in regimes 3–4. Any departure with $m_L < k$ optimally sets $a_L = 1$ (both $U_L$ and $\Pi_{\kappa_L}$ increase in $a_L$ when $c_Q^{eff} < \Lambda$). Along the boundary, $\Pi_{\kappa_H}$ is affine in the promised utility $U_L$ with constant slope $\mu_A(s-\tilde{\rho})/s$, because $(1 - a_H)(k - m_H) = [U(m_H, 1; \kappa_L) - U_L]/s$ and $m_H^*$ does not depend on $U_L$. Meanwhile $U(m_L, 1; \kappa_L)$ and $\Pi_{\kappa_L}(m_L, 1)$ are concave in $m_L$. The leader's objective is therefore concave in $m_L$, and the first-order comparison at $m_L = k$ is necessary and sufficient. Hence, when $\kappa_L \leq \bar{\kappa}_L$, $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is the second-best optimum; when $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$, the leader departs from $(k, 0)$ by setting $a_L^{SB} = 1$ and $m_L^{SB} < k$, establishing claim 5.

In the effort-distortion regime, the shadow value of relaxing $\text{IC}_L$ is different, because the leader uses the extra slack to raise $m_H$ along $a_H = 1$ rather than to raise $a_H$. Raising $U_L$ by one unit raises $m_H$ by $1/(\kappa_L k - c_Q)$ (the slope of $U(\cdot, 1; \kappa_L)$ at $\hat{m}$), and each unit of $m_H$ is worth $\mu_A(c_Q^{eff} - \kappa_H \hat{m}) > 0$ to the leader. Repeating the marginal comparison gives the saturation condition $f_H(c_Q^{eff} - \kappa_H \hat{m}) \leq f_L(c_Q^{eff} - \kappa_L k)$ stated in claim 2. Here the leader's value is no longer affine in $U_L$, so this first-order condition is necessary. An exact grid solver (`verify_prop7_regimes.py`) confirms that it is also sufficient: in all 82 of 120 random effort-regime draws in which it holds, no departure is profitable. The same solver matches the closed forms of regimes 1–4 in all 201 saturated draws of a 270-draw audit.

**Step 5: Comparative statics across the cost spectrum.** In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$ with $k > k_{\text{crit}}$ and $k \geq m_H^*$):

$$1 - a_H^{SB} \;=\; \frac{\frac{\kappa_L(k+m_H^*)}{2} - c_Q}{s}.$$

As $\kappa_L$ increases toward $\bar{\kappa}_L$, type $L$'s utility from completing $k$ attributes decreases, intensifying the temptation to mimic the asking bundle. To deter mimicry, the leader must distort $a_H^{SB}$ progressively downward (lower $a_H$). In the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the leader relieves this pressure by granting clarification to type $L$ ($a_L = 1, m_L < k$), which raises $U_L$ and moderates the downward distortion on type $H$ relative to the saturation formula, dampening distortion growth without reversing its negative direction. Finally, when friction reaches or exceeds task stakes ($c_Q^{eff} \geq \Lambda$, or $\lambda_A - c_Q \geq \mu_A(\Lambda - c_Q)$), the leader prefers zero asking for all types ($\partial\Pi/\partial a = \mu_A(\Lambda - c_Q^{eff})(k - m) \leq 0$). Clarification is uniformly shut down: $a_L = a_H = 0$, and each type is left at its unassisted optimum $m_\kappa = \min\{k, \Lambda/\kappa\}$, which both parties then rank identically. When both types satisfy $\kappa \leq \Lambda/k$, they receive the identical bundle $(k, 0)$ and $\text{IC}_L$ and $\text{IC}_H$ hold with equality. This co-activity reflects technological clarification shutdown due to prohibitive friction, whereas genuine adverse-selection screening distortion ($a_H^{SB} < a_H^B = 1$) operates strictly in the cost-effective asking regime $c_Q < c_Q^{eff} < \Lambda$. □

**Remark 8** (Active Constraint Regimes and Effort versus Asking Distortions)**.** *In classical transfer-based screening (such as Baron–Myerson), the efficient type's incentive constraint ($IC_L$ in our labeling) and the inefficient type's participation constraint ($IR_H$) bind: the efficient type earns an information rent and the inefficient type is distorted and held to its outside option. Here, in the saturation regimes, $IC_L$ and $IR_L$ bind (against the endogenous outside option) while $IR_H$ is slack: the efficient specifier earns no rent, and the distorted high-cost type keeps a rent. Without transfers, the leader cannot reward type L for not mimicking, because asking service is worthless to a user who specifies everything; it can only make the mimicked bundle less attractive. Solving the full four-variable constrained problem reveals five distinct operational regimes:*

1. Unconstrained slack regime ($k \leq k_{\text{slack}}$): *When task complexity is small, the temptation of type L to mimic is absent ($\kappa_L(k+m_H^B) \leq 2c_Q$), so $IC_L$ and $IC_H$ are strictly slack at the unconstrained menu.*

2. Effort-distortion regime ($k_{\text{slack}} < k \leq k_{\text{crit}}$): *Type L faces a temptation to mimic, but the assistant absorbs the entire distortion through specification effort ($m_H = \frac{2c_Q}{\kappa_L} - k < m_H^B$) while preserving full clarification service ($a_H = 1$). Here $IC_L$ binds while asking is undistorted.*

3. Adverse-selection asking distortion regime ($k > k_{\text{crit}}$ with $k \geq m_H^*$): *Specification distortion alone is insufficient, forcing the assistant to distort asking downward ($a_H^{SB} < 1$) alongside effort $m_H = m_H^*$.*

4. Full-specification regime ($k_{\text{slack}} < k < m_H^*$): *Narrow cost heterogeneity forces the interior effort candidate above task complexity ($m_H^* > k$), causing the physical ceiling $m_H \leq k$ to bind; both types specify fully and the menu is payoff-equivalent to pooling at $(k, 0)$ (unless boundary relaxation applies).*

5. Prohibitive friction regime ($c_Q^{eff} \geq \Lambda$, equivalently $\lambda_A - c_Q \geq \mu_A(\Lambda - c_Q)$): *Asking friction exceeds the entire task risk resolved, so the leader prefers zero asking for all types. Clarification is uniformly shut down ($a_L = a_H = 0$) and each type stays at its unassisted optimum $\min\{k, \Lambda/\kappa\}$; when both types saturate at k, they receive the identical bundle $(k, 0)$ and $IC_L$ and $IC_H$ hold with equality trivially. This co-activity reflects technological shutdown, not an adverse-selection screening distortion.*

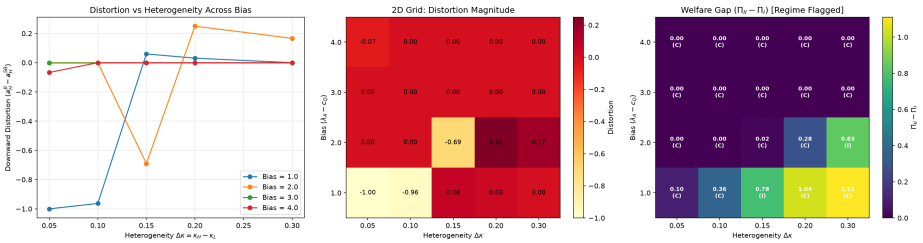



Figure 4: Screening menu distortion across cost heterogeneity $(\kappa_L, \kappa_H)$ and provider friction $\lambda_A$, illustrating adverse-selection downward distortion $a_H^{SB} < 1$.

**Remark 9** (Directionality of Policy Misalignment)**.** *The downward distortion in Proposition 7 is specific to the under-asking bias case ($0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$), which reflects the empirical reality of turn-penalizing satisfaction metrics. Under an over-asking bias ($\lambda_A < c_Q$), the direction of distortion can reverse, as the leader would favor excessive clarification.*

# 7 Comparing the Two Regimes

**Corollary 5** (Menus weakly dominate pooling)**.** *The leader's optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy $a$ is replicated by the menu that offers each type its own best response at ask rate $a$, $\{(m^*(\kappa_L; a), a), (m^*(\kappa_H; a), a)\}$. This menu is incentive compatible, because each type's bundle maximizes its utility among all bundles with ask rate $a$, so the optimal menu is at least as good.*

Beyond payoff dominance, the central comparison lies in the structural decomposition of welfare losses across the two regimes, enabled by Propositions 4–7:

- **Model I (pooling):** Under the linear-risk specification on the physically feasible domain $m \in [0, k]$, an unbiased assistant selects $a^{SE} = 1$, which implements the first-best policy $a^{FB}(\kappa) = 1$ for every type in the population (Proposition 3). Consequently, in the absence of bias ($\lambda_A = c_Q$), the pooling restriction incurs zero welfare loss ($\Pi_{II} - \Pi_I = 0$). Welfare loss under pooling arises when objective bias is present ($\lambda_A > c_Q$), which suppresses the single ask rate $a^{SE}$ below the socially efficient level (Corollary 2), or in the exact conjunctive model (Section 10), where first-best policies differ across types ($a^{FB} \in \{0, 1\}$) and a uniform rate cannot simultaneously tailor to heterogeneous needs.
- **Model II (screening)** incurs welfare loss from bias *alone* (Proposition 6): in the absence of bias, screening recovers full efficiency despite private cost information. Under bias, the loss takes the form of the downward distortion characterized in Proposition 7. This represents a fundamentally different mechanism from the instrument-restriction loss of pooling: it stems not from an inability to condition on observed effort, but from the requirement to prevent opportunistic mimicking across types.

## 7.1 Sensitivity of the Unbiased Pooling Corner and Physical Feasibility

The qualitative finding of Corollary 4, that unbiased pooling is a corner solution rather than an interior compromise, holds for every type distribution, with or without clipping (by the convexity argument given there). The unclipped corner difference formula $\Pi(1) - \Pi(0) = (\Lambda - c_Q)[k - k^*]$ relies

on unconstrained best responses. Table 1 documents this unconstrained boundary across varying task sizes $k$ and baseline guessing accuracies $g$, holding $L = 10, c_Q = 2$, and $\kappa \sim \text{Uniform}[0.2, 0.6]$ (where $\mathbb{E}[1/\kappa] \approx 2.747$).

| $g$ | $\Lambda = L(1-g)$ | Threshold $k^*$ | Task Size $k$ | Unclipped $\Delta\Pi$ | Clipped $\Delta\Pi$ |
|---|---|---|---|---|---|
| 0.30 | 7.0 | 12.36 | 5 | −36.80 | +0.15 |
| 0.30 | 7.0 | 12.36 | 10 | −11.80 | +5.49 |
| 0.30 | 7.0 | 12.36 | 15 | +13.20 | +20.10 |
| 0.50 | 5.0 | 9.61 | 5 | −13.84 | +0.15 |
| 0.50 | 5.0 | 9.61 | 9 | −1.84 | +3.69 |
| 0.50 | 5.0 | 9.61 | 10 (Benchmark) | +1.16 | +5.42 |
| 0.50 | 5.0 | 9.61 | 12 | +7.16 | +9.62 |
| 0.50 | 5.0 | 9.61 | 15 | +16.16 | +17.12 |
| 0.70 | 3.0 | 6.87 | 5 | −1.87 | +0.15 |
| 0.70 | 3.0 | 6.87 | 8 | +1.13 | +1.71 |
| 0.70 | 3.0 | 6.87 | 10 | +3.13 | +3.32 |

Table 1: Sensitivity of the unbiased pooling corner choice across task complexity $k$ and guessing capability $g$, for $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$, where $\Delta\Pi \equiv \Pi(1) - \Pi(0)$. Columns compare the unconstrained closed form $(\Lambda - c_Q)[k - k^*]$ against the physically feasible clipped payoff difference computed via vectorized numerical integration (`underspec_sim/verifications/generate_table1.py`, recorded in `outputs/table1_unbiased_sensitivity.csv`). In the unclipped algebra, guessing appears to win whenever $k < k^*$ (e.g., −36.80 at $g = 0.30, k = 5$), but when followers are constrained to feasible effort $m \leq k$, asking strictly dominates across all rows (+0.15 at $g = 0.30, k = 5$), confirming Proposition 3. While binary symmetric attributes naturally correspond to baseline guessing accuracy $g = 0.50$, configurations with $g = 0.30$ and $g = 0.70$ represent tasks with non-binary attribute domains (such as multi-choice syntactical options where random guessing yields $1/M < 0.5$, or informed priors where guessing yields $> 0.5$).

As shown in Table 1, once followers are restricted to feasible specification effort $m \in [0, k]$, asking ($a^{SE} = 1$) weakly dominates guessing across all 11 configurations under linear risk. The apparent dominance of guessing in the unclipped column stems entirely from low-cost types choosing unfeasible specification levels $m = \Lambda/\kappa > k$. At $k = 5$, for all $\kappa \in [0.2, 0.6]$, the unassisted specification choice under guessing is $\min\{5, \Lambda/\kappa\} = 5 = k$ across all $g \in \{0.30, 0.50, 0.70\}$, because $\Lambda \geq 3.0$ and $\kappa \leq 0.6$. When followers fully saturate specification effort under silence, the task risk parameter $\Lambda$ drops out completely from the comparison, and the net gain from asking $\Delta\Pi = \mathbb{E}[U(m^*(1); 1) - U(k; 0)] = +0.15$ depends strictly on $c_Q = 2$ and $\kappa$. In contrast, when evaluated under the exact conjunctive success function $q(a, g)^{k-m}$, non-linear probability curvature generates genuine interior trade-offs where guessing beats asking even on the interior domain (Section 10).

**Remark 10** (Welfare vs Leader Payoff Ordering Across Bias Bands)**.** *While screening weakly dominates pooling in leader payoff ($\Pi_{II} \geq \Pi_I$) by revealed preference (Corollary 5), the ranking of user welfare $W \equiv \mathbb{E}[U]$ depends non-monotonically on provider asking friction $\lambda_A$ (evaluated at $\kappa_L = 0.3, \kappa_H = 0.5, f_L = 0.5, k = 10, L = 10, g = 0.5, c_Q = 2, V = 100$, and $\mu_A = 1.0$, verified in `underspec_sim/verifications/verify_welfare_ordering.py`):*

1. Mild bias ($c_Q < \lambda_A \leq 3.0$): *Pooling maintains universal clarification ($a^{SE} = 1$) and achieves strictly higher user welfare than screening ($W_{pool} = 85.33$ vs $W_{screen} = 85.00$ at $\lambda_A = 2.5$, gap $-0.33$; $W_{pool} = 85.33$ vs $W_{screen} = 84.52$ at $\lambda_A = 3.0$, gap $-0.81$). The welfare loss under*

*screening at $\lambda_A = 2.5$ arises entirely through* effort dictation. *Here $\kappa_L = 0.3 > c_Q^{eff}/k = 0.25$, so both types lie outside the hypotheses of Proposition 7 and receive the leader's interior preferred bundles $(c_Q^{eff}/\kappa, 1)$, which are incentive compatible. Asking is undistorted ($a_L = a_H = 1$), but the biased assistant demands more upstream specification than users would choose ($m_L = 8.33 > 6.67$, $m_H = 5.00 > 4.00$) to economize on its own query costs. At $\lambda_A = 3.0$ ($\kappa_L = c_Q^{eff}/k$), $IC_L$ binds and the optimal menu is $(m_L, a_L) = (8.67, 1)$, $(m_H, a_H) = (5.56, 0.97)$: effort dictation is now accompanied by a small downward asking distortion on type $H$, further depressing user welfare.*

2. Intermediate bias ($3.0 < \lambda_A < 3.5$): *Pooling drops to $a^{SE} = 2/3$, the kink at which type $L$'s best response leaves the boundary $m = k$, and user welfare falls to $W_{pool} = 82.00$, while screening still delivers $W_{screen} \in (83.0, 83.6)$ (gap $+1.57$ at $\lambda_A = 3.25$). The welfare ranking reverses at $\lambda_A = 3.0$, exactly where pooling stops asking universally.*

3. Moderate bias ($3.5 \leq \lambda_A < 5.0$): *Pooling collapses to zero asking ($a^{SE} = 0$, yielding baseline unassisted welfare $W_{pool} = 80.00$), whereas screening continues to provide partial clarification to terse users ($a_H^{SB} \in [0.70, 0.81]$), delivering strictly higher user welfare ($W_{screen} = 82.45$ at $\lambda_A = 3.5$, gap $+2.45$; 81.53 at $\lambda_A = 4.0$, gap $+1.53$; 80.71 at $\lambda_A = 4.5$, gap $+0.71$).*

4. Prohibitive friction ($\lambda_A \geq 5.0$, i.e., $c_Q^{eff} \geq \Lambda$): *Clarification is priced out under both regimes, so user welfare ties at $W_{pool} = W_{screen} = 80.00$ (gap 0.00).*

*The pooling breakpoints $\lambda_A = 3.0$ and 3.5 are exact. For $a \leq 2/3$, type $L$ is clipped at $m = k$ and type $H$'s contribution is $3(3-2b)a^2$ plus a constant, which is convex for $b < 1.5$ and flat at $b = 1.5$ ($\lambda_A = 3.5$). In the exact conjunctive model, an additional structural instrument-restriction gap appears even at zero bias, separating technological restriction from behavioral misalignment.*

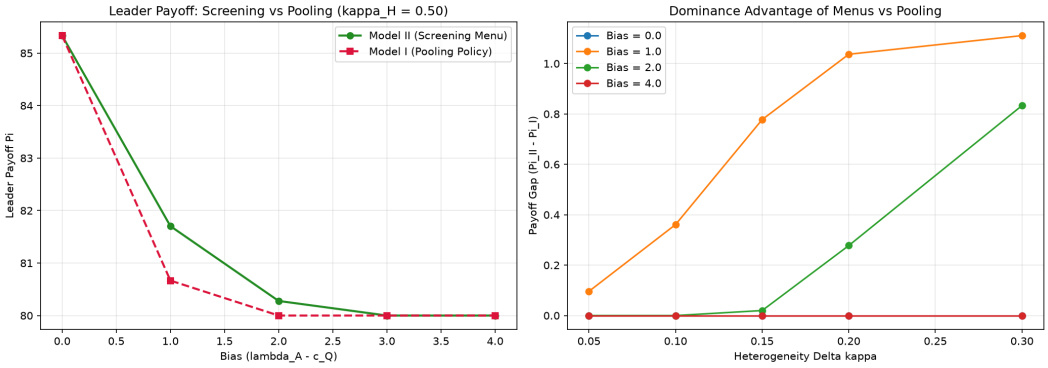



Figure 5: Leader payoff under Model II (screening menu) versus Model I (pooling policy) across assistant bias $\lambda_A - c_Q$ and type heterogeneity $\kappa_H - \kappa_L$, illustrating weak dominance ($\Pi_{II} \geq \Pi_I$) and the zero-gap baseline at zero bias under linear risk.

## 8 Testable Predictions

**P1** (Prop. 1, Cor. 1) Holding the assistant's policy fixed, higher guessing accuracy is associated with shorter specifications alongside weakly higher user welfare, and never with an overall welfare decline despite the reduction in prompt length.

**P1′** (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length flips sign depending on whether $c_Q \gtrless \Lambda = L(1-g)$: where clarifying questions are inexpensive relative to the risk they resolve, users specify less as the ask rate rises (substitution); where questions are relatively costly to answer, users specify *more* (pre-emption). This prediction operates within the model and is distinct from P1's accuracy channel.

**P2** (Prop. 2) Facing the same under-asking assistant, users who overestimate its ask rate (for example, users migrating from a high-asking assistant, or users in an opaque-policy condition) write shorter specifications and attain lower realized task success than users with accurate beliefs. The specification gap scales with $s\,(a^\dagger - a_A)/\kappa$, and, holding policies fixed, it shrinks as guessing accuracy $g$ rises.

**P3** Making ask-rate statistics visible to users shifts specification length toward the sophisticated best response analyzed in Section 3.3.

**P4** (Prop. 7) Ask rates should rise with a user's revealed cost of providing detail (terseness) when clarification policies are screening menus or under nonlinear success models with attribute heterogeneity, rather than responding exclusively to model confidence.

**P5** Users with lower self-reported error-catching rates exhibit declining specification effort across an interactive session (following Section 3's dynamic channel).

**P6** (Cor. 5, Remark 10) Under an assistant with asking friction bias ($\lambda_A > c_Q$), a menu-screening policy achieves weakly higher leader payoff than pooling ($\Pi_{II} \geq \Pi_I$) by tailoring specification requirements to user capabilities. For user welfare ($W \equiv \mathbb{E}[U]$), however, the ranking depends non-monotonically on friction. Under mild bias, while pooling still asks universally ($a^{SE} = 1$; $\lambda_A \leq 3.0$ in the calibration of Remark 10), pooling delivers higher user welfare than screening, which suppresses user surplus through inflated effort requirements and, once $\mathrm{IC}_L$ binds, downward asking distortion. Once friction pushes the pooling rate below one ($3.0 < \lambda_A < 3.5$, $a^{SE} = 2/3$) and then to zero ($3.5 \leq \lambda_A < 5.0$), screening preserves partial clarification for high-cost users and delivers higher user welfare. Under prohibitive friction ($c_Q^{eff} \geq \Lambda$) both regimes shut down asking and tie. In settings characterized by nonlinear exact success or heterogeneous attribute stakes (where first best prescribes asking only for terse users), an unbiased pooling policy also incurs an instrument-restriction loss by forcing uniform asking.

**P7** (Prop. 7) Even when an assistant can condition on declared specification effort, terse users receive fewer clarifying questions than the assistant's unconstrained per-type preference would prescribe ($a_H^{SB} < a_H^B$). This downward distortion arises specifically from the incentive-compatibility constraint $\mathrm{IC}_L$. In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$), downward distortion intensifies as cost heterogeneity narrows ($\kappa_L$ approaches $\bar{\kappa}_L$), requiring deeper distortion to prevent mimicry; in the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the assistant grants clarification to low-cost users ($a_L^{SB} = 1$), relaxing $\mathrm{IC}_L$ and strictly dampening downward distortion on high-cost users relative to the saturated formula (preventing the steep collapse that would occur if $a_L$ were held at 0).

**P8** (Prop. 5) Under an assistant with under-asking friction bias ($\lambda_A > c_Q$), higher model guessing capability $g$ can perversely lengthen user prompts ($d\mathbb{E}[m^*]/dg > 0$) and lower user welfare ($dW/dg < 0$). This equilibrium capability reversal occurs if and only if the assistant's ask-rate reduction is sufficiently steep to outpace direct guessing gains ($-\frac{da^{SE}}{dg} > \frac{L(1-a^{SE})}{\Lambda(g)-c_Q}$), forcing users to over-specify upstream to defensively insure against guessing errors.

# 9 Empirical Strategy and Experimental Design

The behavioral predictions P1–P8 (including P1′) translate our contract-theoretic mechanisms into observable margins for experimental testing. In naturalistic software engineering logs, however, assistant policy bias $\lambda_A$ and user specification costs $\kappa$ are typically confounded with task difficulty and user experience. To causally test our predictions, we propose a between-subjects randomized controlled trial on software developers performing standardized implementation tasks:

**Experimental design and treatments.** Participants are randomly assigned to interact with code assistants operating under distinct clarification regimes:

1. *Treatment A (Low-asking policy, $a = 0.1$):* The assistant aggressively guesses ambiguous attributes with minimal clarification.
2. *Treatment B (High-asking policy, $a = 0.8$):* The assistant frequently asks targeted clarifying questions before generating code.
3. *Treatment C (Menu-conditioned screening policy):* The assistant dynamically selects its clarification rate based on the structural completeness of the initial user prompt, implementing the screening menu derived in Section 6.
4. *Control (Unbiased benchmark):* The assistant asks about every residual ambiguity it detects ($a = 1$), the first-best policy of Proposition 3 under linear risk.

Crossed with the policy arms, a second randomized factor varies *policy disclosure*. In the opaque condition, participants are told nothing about the assistant's asking behavior; in the transparent condition, they see the arm's realized ask rate before the first task. This $4 \times 2$ design separates the belief channel of Proposition 2 (P2, P3) from the substitution channel (P1, P1′). Within the low-asking arm, it also identifies naive under-specification as the opaque–transparent difference in prompt length and task success.

**Measurement, identification, and power analysis.** For each programming session, instrumentation logs: (i) upstream prompt length and specified invariants/edge cases ($m$); (ii) clarification turn count ($a(k - m)$); (iii) downstream test suite execution and correctness; and (iv) post-session subjective developer satisfaction. By exogenously varying the clarification policy $a$, this protocol directly tests whether users economize on prompt specification in response to anticipated assistance ($m^*(a)$ effort substitution, Prediction P1), distinguishes the substitution channel ($\Lambda > c_Q$) from the pre-emption channel ($c_Q > \Lambda$, Prediction P1′), and evaluates whether clarification responds to user specification terseness (Prediction P4).

To identify user specification cost parameters $\kappa$ without confounding from unobserved task difficulty, the protocol includes a standardized pre-trial onboarding task where participants articulate requirements without assistant interaction, establishing an individual-level baseline specification effort per unit of task complexity. Assistant asking friction $\lambda_A$ is exogenously manipulated through provider system prompt instructions, conversational turn penalties, and backend latency budgets. A power calculation for the primary contrast (prompt length $m$, low-asking versus high-asking arm, two-sided two-sample $t$-test) shows that detecting a small-to-medium effect of $d = 0.35$ standard deviations at $\alpha = 0.05$ with 80% power requires $n = 130$ developers per policy arm, or 520 across the four arms. If all six pairwise arm contrasts are treated as confirmatory with a Bonferroni correction ($\alpha = 0.05/6$), the requirement rises to $n = 200$ per arm. Because the disclosure factor is crossed within arms, its main effect is estimated on the full sample; the opaque–transparent

contrast within the low-asking arm alone has power 0.51 at $d = 0.35$ with $n = 130$ per arm (about 65 per cell), so testing P2/P3 within that arm requires either a larger arm or a pre-registered one-sided test. Because each developer completes several tasks, within-subject repetition and task fixed effects further reduce residual variance. We therefore recommend pre-registering the primary contrast, the $d$ assumption, and the correction method.

**Ethics and human subjects considerations.** The proposed human evaluation involves non-invasive coding tasks. Participating developers should be recruited through professional developer platforms, provide explicit informed consent, and receive compensation at standard market rates for professional software engineering (e.g., \$50–\$75 per hour). All logged code, prompts, and telemetry must be fully anonymized prior to analysis, with no personally identifiable information recorded. The study protocol should be submitted to the relevant Institutional Review Board (IRB) for formal exemption or expedited review under institutional guidelines for research involving benign behavioral interventions. To facilitate reproducibility and protocol validation prior to live human deployment, we provide a companion open-source simulation harness (`underspec_sim/experiments_llm`) supporting both deterministic offline dry-run evaluation with mock LLM client responses and live API interactions across pooling and screening treatments.

## 10 Discussion and Limitations

The linear-risk formulation is the main tractability baseline analyzed in Sections 4–7. Rather than viewing linear risk solely as a small-risk Taylor approximation, it is properly understood as an independent, additive-defect primitive model:

$$U(m, a; \kappa) \;=\; V \;-\; L(1-q)(k-m) \;-\; \frac{\kappa}{2}m^2 \;-\; a\, c_Q\,(k-m),$$

where $L$ represents the localized defect cost per unresolved attribute and cumulative defect loss satisfies $kL \leq V$. In this additive environment, each unaddressed attribute produces an independent defect without destroying the entire project. By contrast, the exact conjunctive specification $Vq(a, g)^{k-m}$ represents a systemic bottleneck environment where all attributes are strictly complementary and a single error destroys the entire deliverable. To calibrate the exact model against the linear baseline, note that for benchmark task size $k = 10$, total project valuation $V = 100$ corresponds precisely to cumulative defect loss $kL = 100$ (with per-attribute stake $L = 10$). To investigate the role of payoff curvature, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a, g)^{k-m}$ via direct numerical grid search and constrained optimization (implemented in `verify_exact_conjunctive.py`, recorded in `outputs/exact_conjunctive_robustness.csv`). Rather than merely replicating the linear model, the exact conjunctive specification represents a distinct, richer economic environment where probability curvature introduces genuine interior trade-offs.

Under this exact nonlinear specification across task complexity $k \in [3, 20]$ and guessing accuracy $g \in [0.50, 0.95]$ (30 grid points):

1. The corner-solution property of Corollary 4 persists universally ($a^{SE}_{\text{exact}} \in \{0, 1\}$ in all 30 configurations).

2. Unbiased pooling choices agree between the exact model and the linear Taylor-expansion baseline ($L = V = 100$) in 25 of 30 configurations (83.3%), with all 5 divergences concentrated exclusively at small $k = 3$ where compounding ($q^3$) makes autonomous guessing preferred over question

friction. For all $k \geq 5$ across all $g \in [0.50, 0.95]$, both models select universal clarification $a^{SE} = 1.0$. Under the localized additive-defect baseline ($L = 10$), agreement is 17 of 30 configurations (56.7%), with divergences at high $g \geq 0.90$ where localized attribute stakes make guessing cost-effective earlier than systemic failure stakes. This nonlinear curvature creates a positive instrument-restriction loss at zero bias in the exact model that is absent in linear risk.

3. Downward screening distortion ($a_H^{SB} < a_H^B$) holds in 20 of 30 cases (67% overall: 20% at $k \in \{3, 5\}$, 80% at $k \in \{8, 20\}$, and 100% at $k \in \{10, 15\}$). The low incidence of distortion at small $k$ stems from boundary saturation: when tasks are simple ($k \leq 5$), users can fully specify all attributes ($m = k$), leaving no residual ambiguity for clarification to resolve ($k - m = 0$). Consequently, both $\text{IC}_L$ and $\text{IC}_H$ are active in 15 of 30 cases, concentrated entirely at small task sizes ($k \leq 8$).

Furthermore, numerical solutions of the full four-variable constrained menu problem under the exact payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ (`verify_exact_bias_sweep.py`, recorded in `outputs/exact_bias_sweep.csv` and Figure 6) illustrate the boundary transition to clarification shutdown. In the linear model with $\mu_A = 1.0$, asking friction reaches or exceeds task stakes ($c_Q^{eff} \geq \Lambda$) at $\lambda_A \geq 5.0$. Beyond this threshold, the leader unconstrainedly prefers zero asking for all types ($a_L^B = a_H^B = 0$). Under the exact nonlinear payoff, this technological clarification shutdown threshold shifts outward to $\lambda_A \approx 8.0$. In this shutdown regime, both types choose full specification $m = k$ and receive $(k, 0)$, so $\text{IC}_H$ and $\text{IC}_L$ both hold with equality ($0 = 0$). This co-activity is not an adverse-selection screening distortion, but a technological shutdown artifact where clarification is universally priced out by severe asking friction.

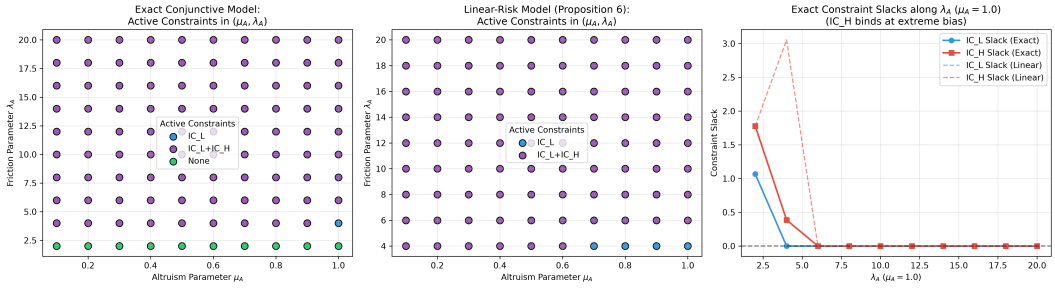



Figure 6: Screening menu allocations and active constraint boundaries under the exact conjunctive payoff across assistant asking friction $\lambda_A$ and weight $\mu_A$, illustrating the transition to clarification shutdown at $\lambda_A \approx 8.0$.

Another limitation concerns the off-menu contract schedule. Proposition 7 is implemented by a discontinuous policy that asks at rate $a_H^{SB}$ only at the menu specification level and does not ask ($a = 0$) at any other level. Lemma 2 shows that this schedule is incentive compatible even when users may type any prompt, so the menu does not rely on users being restricted to the two listed bundles. Still, a schedule that withholds clarification from a prompt slightly longer or shorter than $m_H^{SB}$ may be brittle in practice, because specification levels are measured with noise, or may seem arbitrary to users. Exploring smooth, continuous UI policies $a(m)$ or graded clarification schedules that remain robust to prompt gaming offers a fruitful avenue for applied mechanism design.

Regarding the regularity condition (Assumption 1), decreasing differences on the entire domain $m \in [0, k]$ (the hypothesis of Topkis's theorem) reduces to $k \ln q \geq -1$, which holds in 24.8% of a broad numerical parameter grid. For local comparative statics at a smooth interior optimum, the sign condition need only hold at $m^*$ (Remark 1). The local condition $(k - m^*) \ln q \geq -1$ holds

in 89.9% of the grid (`verify_assumption1.py`). A finite-difference audit (`verify_monotonicity_survival.py`, recorded in `outputs/monotonicity_survival.csv`) finds that $m^*(g)$ is weakly decreasing in all 2,162 grid intervals where the local condition holds. In the 390 intervals where it fails, monotonicity breaks in 377 (96.7%): higher guessing accuracy raises specification effort, and the 13 survivors are boundary-saturated at $m^* = k$. The local condition is therefore close to necessary as well as sufficient. Under the exact conjunctive payoff, Corollary 1 should be read as a statement about the empirically relevant region where users leave few attributes unspecified, not as a global law.

A stylized modeling assumption is attribute symmetry ($k$ binary attributes with identical loss $L$ and guessing probability $g$). In practical software engineering, attributes exhibit substantial loss heterogeneity: omitting an architectural invariant or security convention causes severe failure ($L_i \gg 0$), while omitting a stylistic naming convention carries negligible cost ($L_j \approx 0$). In an extension with attribute-specific losses $L_i$, the user's objective becomes $U = V \prod_i q_i - c(m) - \sum_i a_i c_Q$. Ordering attributes by descending risk stake $\Lambda_i = L_i(1 - g_i)$, users rationally specify the highest-stakes attributes first, while an efficient assistant asks only on residual attributes with $\Lambda_i > c_Q$. We conjecture that the core mechanism, the substitution between upstream specification effort and automated clarification under misaligned asking friction, carries over to this setting; a formal treatment must handle the endogenous ordering of which attributes are specified and which are left to the assistant, and is left for future work.

A related limitation concerns the empirical identification of $\lambda_A$. Every bias-driven prediction (Corollary 2, Proposition 7, and predictions P2, P6, P7, and P8) is stated in terms of under-asking bias ($\lambda_A > c_Q$), but backing out $\lambda_A$ from observed policy alone is circular. Lemma 1 formalizes this as an exact non-identification result: $a^{SE}$ depends on $(\mu_A, \lambda_A)$ only through their ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$, so a discounting-driven leader and a friction-driven leader with the identical excess friction ratio cannot be distinguished from pooling behavior alone. An independent empirical handle, such as an explicit penalty on turn count in the training objective or reward specification, would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested without circularity. The two-type restriction in Model II avoids ironing complications at the cost of generality; a continuum formulation using the standard Mirrlees technique (Mirrlees, 1971; Myerson, 1979) is a natural next step to characterize the continuous distribution of downward distortions across arbitrary cost distributions. We also assume truthful revelation of specified attributes (abstracting from cheap-talk ambiguity within what is communicated) and a single leader. Modeling competition among multiple assistants represents an appealing extension, as market competition could plausibly discipline asking bias directly.

A broader methodological conclusion concerns the limitations of applying classical contract-theoretic intuition directly to human–AI settings without accounting for non-monetary, partially aligned incentives. In classical mechanism design (such as Baron–Myerson, Laffont–Tirole, or costly verification models (Baron and Myerson, 1982; Laffont and Tirole, 1986; Ben-Porath et al., 2014)), the principal pays monetary transfers to extract surplus, and agent informational rent directly reduces the principal's payoff. In human–AI interaction, the assistant's objective directly incorporates user welfare. As a consequence, incentive compatibility is costless in the absence of bias (Proposition 6). Under bias, the rent pattern is reversed: the efficient specifier is held to its outside option while the distorted high-cost type retains a rent (Remark 8). And unbiased pooling induces corner solutions rather than interior compromises (Corollary 4). Economic analyses of automated assistance must therefore explicitly incorporate the non-monetary, partially aligned nature of the interaction rather than relying on direct analogies to cash-transfer mechanisms.

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